

3

Project Management

LEARNING OBJECTIVES

Once you have mastered the material in this chapter, you will be able to:

1. Understand how projects are initiated and selected, define a business problem, and determine the feasibility of a proposed project.
2. Evaluate hardware and software alternatives by addressing the trade-offs.
3. Forecast and analyze tangible and intangible costs and benefits.
4. Manage a project by preparing a budget, creating a work breakdown structure, scheduling activities, and controlling the schedule and costs.
5. Build and manage a project team.
6. Write an effective systems proposal, concentrating on both content and design.

Initiating projects, determining project feasibility, scheduling projects, estimating costs, budgeting, and planning and then managing activities and team members for productivity are all important capabilities for a systems analyst to master. As such, they are considered project management fundamentals.

A systems project begins with problems or with opportunities for improvement in a business that come up as the organization adapts to change. The increasing popularity of ecommerce means that some fundamental changes are occurring as businesses either originate their enterprises on or move their internal operations and external relationships to the Internet. Changes that require a systems solution occur in the legal environment as well as in the industry's environment. A systems analyst works with users to create a problem definition that reflects current business systems and concerns.

Once a project is suggested, the systems analyst works quickly with decision makers to determine whether the project is feasible. When a project is approved for a full systems study, the project activities are scheduled through the use of tools such as Gantt charts and Program Evaluation and Review Technique (PERT) diagrams so that the project can be completed on time. Part of ensuring the productivity of systems analysis team members is effectively managing their scheduled activities. This chapter is devoted to a discussion of project management fundamentals.

Project Initiation

Systems projects are initiated by many different sources for a variety of reasons. Some of the projects suggested will survive various stages of evaluation to be worked on by you (or you and your team); others will not and should not get that far. Businesspeople suggest systems projects for two broad reasons: (1) because they experience problems that lend themselves to systems solutions, and (2) because they recognize opportunities for improvement through upgrading, altering, or installing new systems. Both situations can arise as an organization adapts to and copes with natural, evolutionary change.

Problems in an Organization

Managers do not like to conceive of their organization as having problems, let alone talk about those problems or share them with someone from outside. Good managers, however, realize that recognizing symptoms of problems or, at a later stage, diagnosing the problems themselves and then confronting them are imperative if the business is to keep functioning at its highest potential.

Problems surface in many different ways. In some instances, problems are uncovered because performance measures are not being met. Problems (or symptoms of problems) with processes that are visible in output and that could require the help of a systems analyst include excessive errors and work performed too slowly, incompletely, incorrectly, or not at all. Other symptoms of problems become evident when people do not meet baseline performance goals. Changes in employee behavior such as unusually high absenteeism, high job dissatisfaction, or high worker turnover should alert managers to potential problems. Any of these changes, alone or in combination, might be a sufficient reason to request the help of a systems analyst.

Although difficulties such as those just described occur in an organization, feedback on how well the organization is meeting intended goals may come from outside, in the form of complaints or suggestions from customers, vendors, or suppliers, as well as lost or unexpectedly low sales. This feedback from the external environment is extremely important and should not be ignored. A summary of symptoms of problems and approaches useful in problem detection is provided in Figure 3.1.

Defining the Problem

Whether using the classic systems development life cycle (SDLC) approach or an object-oriented approach, an analyst first defines the problems and objectives of the system. These form the foundation for determining what needs to be accomplished by the system.

A problem definition usually contains some sort of problem statement, summarized in a paragraph or two. This is followed by a series of issues or major independent pieces of the problem. The issues are followed by a series of objectives or goals that match the issues point by point. Issues are the current situation; objectives are the desired situation. The objectives may be very specific or worded as a general statement.

To Identify Problems	Look for These Specific Signs:
Check output against performance criteria.	• Too many errors • Work completed slowly • Work done incorrectly • Work done incompletely • Work not done at all
Observe behavior of employees.	• High absenteeism • High job dissatisfaction • High job turnover
Listen to external feedback from: Vendors and service providers. Customers. Suppliers.	• Complaints • Suggestions for improvement • Loss of sales • Lower sales

FIGURE 3.1

Checking output, observing employee behavior, and listening to feedback are all ways to help the analyst pinpoint systems problems and opportunities.



CONSULTING OPPORTUNITY 3.1

The Sweetest Sound I've Ever Sipped

Felix Straw, who represents one of the many U.S. distributors of the European soft drink Sipps, gazes unhappily at the weather app on his smartphone, which shows a map of the United States saturated with dark red, indicating that most of the country is experiencing an early spring heat wave, with no signs of a letup. Holding up the screen of his smartphone as he speaks to the assembled systems group, he says, “It’s the best thing that could happen to us, or at least it should be. But when we had to place our orders three months ago, we had no idea that this spring monster heat wave was going to devour the country this way!” Nodding his head toward a picture of their European plant hung on the wall, he continues, “We need to be able to tell them when things are hot over here so we can get enough product. Otherwise, we’ll miss out every time. This happened two years ago, and it just about killed us.”

“Each of us distributors meets with our district managers to do three-month planning. When we agree, we email our orders to European headquarters. They make their own adjustments, bottle the drinks, and then we get our modified orders about 9 to 15 weeks later. But we need ways to tell them what’s going on now. Why, we even have some new superstores that are opening up here. They should know we have extra-high demand.”

Corky, his assistant, agrees, saying, “Yeah, they should at least look at our past sales around this time of year. Some springs are hot, others are just average.”

Straw concurs, saying, “It would be music to my ears, it would be really sweet, if they would work with us to spot trends and changes—and then respond quickly.”

Stern’s, based in Blackpool, England, is a European beverage maker and the developer and producer of Sipps. Sipps is a sweet, fruit-flavored, nonalcoholic, noncarbonated drink served chilled or with ice, and it is particularly popular when the weather is hot. Sipps has sold briskly in Europe and grown in popularity in the United States since its introduction five years ago, but the company has had a difficult time adequately managing inventory and keeping up with U.S. customer demand, which is affected by seasonal temperature fluctuations. Places with year-round, warm-temperature climates and lots of tourists (such as Florida and California) have large standing orders, but other areas of the country could benefit from a less cumbersome, more responsive order-placing process. A network of local distributors located throughout the United States and Canada distributes Sipps.

As one of the systems analysts assigned to work with the U.S. distributors of Sipps, you begin your analysis by listing some of the key symptoms and problems you have identified after studying the information flows, ordering process, and inventory management, and after interviewing Mr. Straw and his assistant. In a paragraph, describe which problems might indicate the need for a systems solution.

Note: This consulting opportunity is loosely based on J. C. Perez, “Heineken’s HOPS Software Keeps A-Head on Inventory,” *PC Week*, 14, 2 (January 13, 1997): 31 and 34.

Here are some examples of business questions relating to business objectives:

- What are the purposes of the organization?
- Is the organization a for-profit or nonprofit organization?
- Does the company plan to grow or expand?
- What is the organization’s attitude (culture) about technology?
- What is the organization’s budget for IT?
- Does the organization’s staff have the expertise?

Needless to say, a systems analyst needs to understand how a business works.

Finally, the problem definition contains requirements—the things that must be accomplished—along with the possible solutions and the constraints that limit the development of the system. The requirements section may include security, usability, government requirements, and so on. Constraints often include the word *not*, indicating a limitation, and may contain budget restrictions or time limitations.

The problem definition is produced after completing interviews, observations, and document analysis with the users. The result of gathering this information is a wealth of facts and important opinions in need of summary. The first step in producing the problem definition is to find a number of points that may be included in one issue. Major points can be identified in the interview in a number of ways:

1. Users may identify an issue, a topic, or a theme that is repeated several times, sometimes by different people in several interviews.
2. Users may communicate using the same metaphors, such as saying the business is a journey, a war, a game, an organism, a machine, a family, and so on.

3. Users may tell a story to illustrate a problem that includes a beginning, a middle, and an ending; a hero; obstacles to overcome; and a successful (or hoped for) resolution.
4. Users may speak at length on a topic.
5. Users may tell you outright, “This is a major problem.”
6. Users may communicate importance using body language or may speak emphatically on an issue.
7. The problem may be the first thing the user mentions.

Once the issues have been identified, the objectives must be stated. An analyst may have to do a follow-up interview to obtain more precise information about the objectives. After the objectives are stated, the relative importance of the issues or objectives must be determined. If there are not enough funds to develop the complete system, the critical objectives must be completed first. Users are the best people to identify critical objectives (with the support of analysts) because users are domain experts in their business area and know how they work best with technologies in the organization. In addition to looking through data and interviewing people, a systems analyst should try to witness the problem firsthand.

A PROBLEM DEFINITION EXAMPLE: CATHERINE'S CATERING Catherine's Catering is a small business that caters meals, receptions, and banquets for business and social occasions such as luncheon meetings and weddings. It was inspired by Catherine's love of cooking and talent for preparing fine meals. At first it was a small company with a handful of employees working on small projects. Catherine met with customers to determine the number of people, the types of meals, and other information necessary to cater an event. As her company's reputation for creating superb food and the quality of the service began to blossom, the number of events started to increase. A new convention center built in the city and a prospering business community increased the number of catering events.

Catherine was able to manage the business using spreadsheets and word processing but had difficulty keeping up with the endless phone calls about what types of meals were available, changes to the number of guests attending the event, and the availability of specialty dietary items, such as vegan, vegetarian, low-fat, low-carbohydrate, gluten-free, and so on. Catherine hired a number of part-time employees to cook and cater the events, however the complexity of scheduling personnel was becoming overwhelming to the new human resources manager. Catherine decided to hire an IT and business consulting company to help her address the problems her catering enterprise was facing.

After performing interviews and observing a number of key staff, the consultants found the following concerns:

1. The master chef ordered supplies (produce, meat, and so on) from suppliers for each event. The suppliers would provide discounts if greater quantities were ordered at a single time for all events occurring in a given time frame.
2. Customers often called to change the number of guests for an event, with some changes made only one or two days before the event was to occur.
3. It was too time-consuming for Catherine and her staff to handle each request for catering, with about 60 percent of the calls resulting in a contract.
4. Conflicts in employee schedules were occurring, and some events were understaffed. Complaints about the timeliness of service were becoming more frequent.
5. Catherine does not have any summary information about the number of events and types of meals. It would be helpful to have trend information to help guide customers in their choice of meals.
6. Events that include service for sit-down meals are often held in banquet rooms and at other meeting halls. There are problems with scheduling sufficient wait staff and responding to changes in the number of guests.

The problem definition is shown in Figure 3.2. Notice the weights on the right, representing an average of the weights assigned by each employee. Objectives match the issues. Each objective is used to create user requirements.

User requirements are then used to create either use cases and a use case diagram or processes on a data flow diagram. Each objective may have one or more user requirements,

FIGURE 3.2

Problem definition for Catherine's Catering developed with the help of users.

Catherine's Catering

Problem Definition

Catherine's Catering is experiencing problems with handling the number of routine calls with customers, as well as coordinating with external partners such as suppliers and meeting facilities. The growth in the number of part-time staff is leading to scheduling conflicts and understaffed events.

Issues

	Weight
1. Customer contact takes an inordinate amount of time for routine questions.	10
2. Managing part-time employees is time consuming and leads to scheduling errors.	9
3. It is difficult to accommodate last-minute changes for events.	7
4. Supplies are ordered for each event. Often shipments are received several times a day.	6
5. There are often problems communicating changes to event facilities.	5
6. There is little historical information about customers and meals.	3

Objectives

1. Provide a Web system for customers to obtain pricing information and place orders.
2. Create or purchase a human resources system with a scheduling component.
3. After customers have signed an event contract, provide them with Web access to their account and a means for them to update the number of guests. Notify management of changes.
4. Provide a means to determine overall quantities of supplies for events occurring within a concurrent time frame.
5. Provide a system for communicating changes to key personnel at event facilities.
6. Store all event data and make summary information available in a variety of formats.

Requirements

1. The system must be secure.
2. Feedback must be entered by event managers at the close of each event.
3. There must be a means for event facilities to change their contact person.
4. The system must be easy to use by nontechnical people.

Constraints

1. Development costs must not exceed \$50,000.
2. The initial website for customer orders must be ready by March 1 to accommodate requests for graduation parties and weddings.

several objectives may lead to the creation of one or perhaps no use cases (use cases are not often created for simple reports), or each requirement may lead to the creation of one data flow diagram process. The user requirements for Catherine's Catering are to:

1. Create a dynamic website to allow current and potential clients to view and obtain pricing information for a variety of different products.
2. Allow current and potential clients to submit a request with their catering choices, with the request routed to an account manager.
3. Add clients to the client database, assigning each a userID and a password for access to their projects.
4. Create a website for clients to view and update the number of guests for an event and restrict changes in the number of guests when the event is less than five days in the future.
5. Obtain or create software to communicate directly with event facility personnel.
6. Develop or purchase a human resources system for scheduling part-time employees that allows management to add employees and schedule them using a number of constraints.
7. Provide queries or reports with summary information.

Each requirement may be used to create a preliminary test plan. Few details are available at this time, so the test plan will be revised as the project progresses.

A simple test plan for Catherine's Catering is:

1. Design test data to allow clients to view each different type of product.
2. Test to ensure that a catering request has been entered with valid data, as well as each possible condition of invalid data. (Data will be defined later.) Ensure that the request is routed to the appropriate account manager.
3. Test that all data fields pass all validation criteria for each field. Test good data to ensure that clients are added to the client database and that a userID and a password are correctly assigned.
4. Create a test plan that will confirm that clients are able to view event information. Test that updates may not be made within five days of the event. Design test data to ensure correct updating of the number of guests for an event.
5. Test that the software works correctly for communicating directly with event facility personnel.
6. Test the human resources system for scheduling part-time employees, checking that employees have been correctly added and that all invalid values for each field are detected and reported. Check scheduling software for valid updates and each invalid entry.
7. Check that all queries and reports work correctly and contain the correct summary information.

Selection of Projects

Projects come from many different sources and for many reasons. Not all should be selected for further study. You must be clear in your own mind about the reasons for recommending a systems study on a project that seems to address a problem or could bring about improvement. Consider the motivation that prompts a proposal on the project. You need to be sure that the project under consideration is not being proposed simply to enhance your own political reputation or power or that of the person or group proposing it, because there is a high probability that such a project will be ill conceived and eventually poorly accepted.

As outlined in Chapter 2, prospective projects need to be examined from a systems perspective, considering the impact of the proposed change on the entire organization. Recall that the various subsystems of an organization are interrelated and interdependent, so a change to one subsystem might affect all the others. Even though the decision makers directly involved ultimately set the boundaries for a systems project, a systems project cannot be contemplated or selected in isolation from the rest of the organization.

Beyond these general considerations are five specific questions that need to be asked regarding project selection:

1. Does it have backing from management?
2. Is the timing of the project commitment appropriate?
3. Is it possible to improve attainment of strategic organization goals?
4. Is it practical in terms of resources for the systems analyst and organization?
5. Is it a worthwhile project compared with other ways the organization could invest resources?

First and foremost is backing from management. Absolutely nothing can be accomplished without the endorsement of the people who will eventually foot the bill. This statement does not mean that you lack influence in directing the project or that people other than management can't be included; however, management backing is essential.

Another important criterion for project selection is timing for you and the organization. Ask yourself and the others who are involved whether the business is presently capable of making a time commitment for installation of new systems or improvement to existing ones. You must also be able to commit all or a portion of your time for the duration.

A third criterion is the possibility of improving attainment of strategic organization goals such as (1) improving corporate profits, (2) supporting the competitive strategy of the organization, (3) improving cooperation with vendors and partners, (4) improving internal

operations support so that goods and services are produced efficiently and effectively, (5) improving internal decision support so that decisions are more effective, (6) improving customer service, and (7) improving employee morale. The project should put the organization on target, not deter it from its ultimate goals.

A fourth criterion is selecting a project that is practicable in terms of your resources and capabilities as well as those of the business. Some projects will not fall within your realm of expertise, and you must be able to recognize them.

Finally, you need to come to a basic agreement with the organization about the worthiness of the systems project relative to any other possible project being considered. There are many possibilities for improvements, including (1) speeding up a process, (2) streamlining a process through the elimination of unnecessary or duplicated steps, (3) combining processes, (4) reducing errors in input through changes of forms and display screens, (5) reducing redundant storage, (6) reducing redundant output, and (7) improving integration of systems and subsystems. Remember that when a business commits to one project, it is committing resources that thereby become unavailable for other projects. It is useful to view all possible projects as competing for the business resources of time, money, and people.

Determining Feasibility

Once the number of projects has been narrowed according to the criteria discussed previously, it is still necessary to determine whether the selected projects are feasible. Our definition of *feasibility* goes much deeper than the common usage of the term. Systems project feasibility is assessed in three principal ways: operationally, technically, and economically. A feasibility study is not a full-blown systems study. Rather, a feasibility study is used to gather broad data for the members of management that enables them to make a decision about whether to proceed with a systems study.

Determining Whether It Is Possible

After an analyst determines reasonable objectives for a project, the analyst needs to determine whether it is possible for the organization and its members to see the project through to completion. Generally, the process of feasibility assessment is effective in screening out projects that are inconsistent with the business’s objectives, that are technically impossible, or that have no economic merit.

Although it is painstaking, studying feasibility is worthwhile because it saves businesses and systems analysts time and money. In order for an analyst to recommend further development, a project must show that it is feasible in all three of the following ways: technically, economically, and operationally (see Figure 3.3).

FIGURE 3.3
 The three key elements of feasibility are technical, economic, and operational feasibility.

The Three Key Elements of Feasibility
Technical Feasibility Add on to present system Technology available to meet users’ needs
Economic Feasibility Systems analysts’ time Cost of systems study Cost of employees’ time for study Estimated cost of hardware Cost of packaged software or software development
Operational Feasibility Whether the system will operate when put in service Whether the system will be used

TECHNICAL FEASIBILITY An analyst must find out whether it is possible to develop a new system given the current technical resources. If not, can the system be upgraded or added to in a manner that fulfills the request under consideration? If existing systems cannot be added to or upgraded, the next question becomes whether there is technology in existence that meets the specifications.

At the same time, the analyst can ask whether the organization has staff who are technically proficient to accomplish the objectives. If not, can the organization hire additional coders, testers, experts, or others who may have different coding skills from theirs, or perhaps outsource the project completely? In addition, are software packages available that can accomplish the objectives, and how significantly does that software need to be customized for the organization?

ECONOMIC FEASIBILITY Economic feasibility is the second part of resource determination. The basic resources to consider are your time and that of the systems analysis team, the cost of doing a full systems study (including the time of employees you will be working with), the cost of the business employee time, the estimated cost of hardware, and the estimated cost of software, software development, or software customization.

The organization must be able to see the value of the investment it is pondering before committing to an entire systems study. If short-term costs are not overshadowed by long-term gains or if the project produces no immediate reduction in operating costs, the system is not economically feasible and should not proceed further.

OPERATIONAL FEASIBILITY Suppose for a moment that technical and economic resources are both judged adequate. The systems analyst must now consider the operational feasibility of the requested project. Operational feasibility is dependent on the human resources available for the project and involves projecting whether the system will operate and be used once it is put into service.

If users are virtually wed to the present system, see no problems with it, and generally are not involved in requesting a new system, resistance to implementing the new system will be strong. The new system has a low chance of becoming operational.

Alternatively, if users themselves have expressed a need for a system that is operational more of the time and is more efficient and more accessible, chances are better that the requested system will eventually be used. Much of the art of determining operational feasibility rests with the user interfaces that are chosen, as we see in Chapter 14.

Estimating Workloads

The next step in ascertaining hardware needs is to estimate workloads. Thus, systems analysts formulate numbers that represent both current and projected workloads for the system so that any hardware obtained will be capable of handling current and future workloads.

If estimates are accurate, the business should not have to replace hardware solely due to unforeseen growth in system use. (Other events, however, such as superior technological innovations, may dictate hardware replacement if the business wants to maintain its competitive edge.)

Out of necessity, workloads are sampled rather than actually put through several computer systems. The guidelines given in Chapter 5 can be of use here because in workload sampling the systems analyst is taking a sample of necessary tasks and the computer resources required to complete them.

Figure 3.4 compares the time required by existing and proposed information systems that are supposed to handle a given workload. Notice that the company is currently using a legacy computer system to prepare a summary of shipments to its distribution warehouses, and a Web-based dashboard is being suggested. The workload comparison looks at when and how each process is done, how much human time is required, and how much computer time is needed. Notice that the newly proposed system should reduce the required human and computer time significantly.

FIGURE 3.4

Comparisons of workloads between existing and proposed systems.

	Existing System	Proposed System
Task	Compare performance of distribution warehouses by running the summary program.	Compare performance of distribution warehouses on the Web-based dashboard.
Method	Computer programs are run when needed; processing is done from the workstation.	Updates occur immediately; processing is done online.
Personnel	Distribution manager	Distribution manager
When and how	Daily: Enter shipments on Excel spreadsheet; verify accuracy of spreadsheet manually; and then write files to backup media. Monthly: Run program that summarizes daily records and prints report; get report and make evaluations.	Daily: Enter shipments on the Web-based system using drop-down boxes. Data are automatically backed up to remote location. Monthly: Compare warehouses online using the performance dashboard; print only if needed.
Human time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes
Computer time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes

Ascertaining Hardware and Software Needs

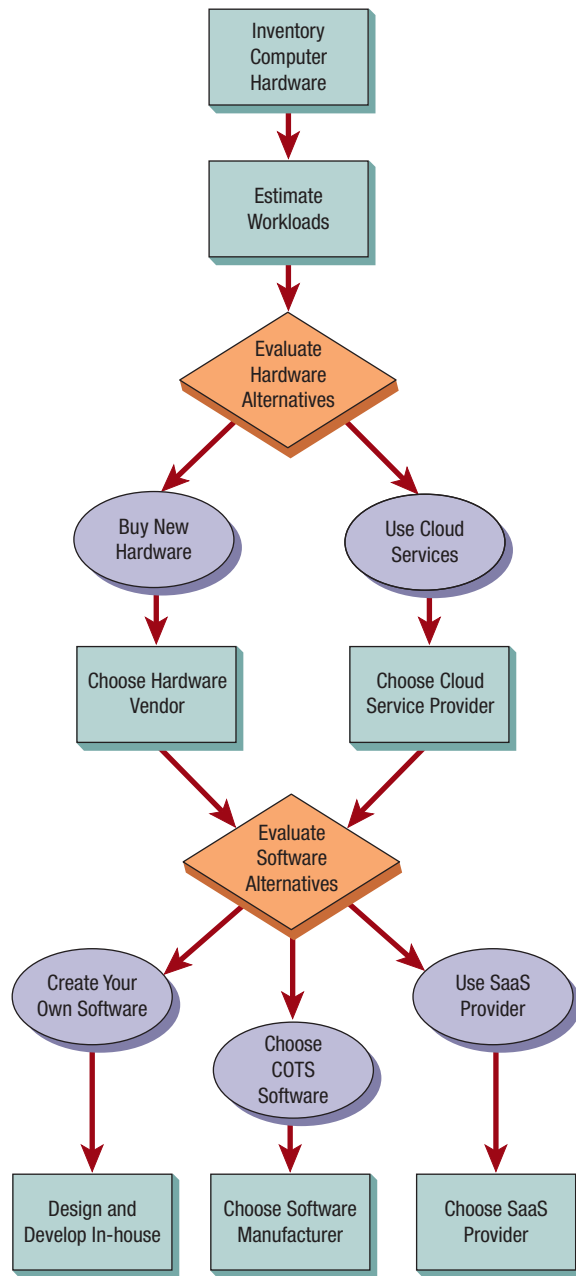
Assessing technical feasibility includes evaluating the ability of computer hardware and software to handle workloads adequately. Figure 3.5 shows the steps a systems analyst takes in ascertaining hardware and software needs. First, all current computer hardware the organization owns must be inventoried to discover what is on hand and what is usable.

A systems analyst needs to work with users to determine what hardware will be needed. Hardware determinations can be made only in conjunction with determining human information requirements. Knowledge of the organization structure and how users interact with technologies in an organization setting also can be helpful in hardware decisions. Only when systems analysts, users, and management have a good grasp of what kinds of tasks must be accomplished can hardware options be considered.

Inventorizing Computer Hardware

A systems analyst needs to inventory what computer hardware is already available in the organization. As will become apparent, some of the hardware options involve expanding or recycling current hardware, so it is important to know what is on hand.

If an updated computer hardware inventory is unavailable, the systems analyst needs to set one up quickly and carry through on it. It is important to know the following:

**FIGURE 3.5**

Steps in choosing hardware and software.

1. The type of equipment, including model number and manufacturer
2. The operational status of the equipment, such as on order, operating, in storage, or in need of repair
3. The estimated age of the equipment
4. The projected life of the equipment
5. The physical location of the equipment
6. The department or person considered responsible for the equipment
7. The financial arrangement for the equipment, such as owned, leased, or rented

Ascertaining the current hardware available will result in a sounder decision-making process when hardware decisions are finally made because much of the guesswork about what exists will have been eliminated. Through your earlier interviews with users, questionnaires surveying them, and research of archival data, you will already know the number of people available for data processing as well as their skills and capabilities. Use this information to project how well the staffing needs for new hardware can be met.

Evaluating Computer Hardware for Purchase

Evaluating new computer hardware is the shared responsibility of management, users, and systems analysts. Although vendors will be supplying details about their particular offerings, analysts need to oversee the evaluation process personally because they will have the best interests of the business at heart. In addition, systems analysts may have to educate users and management about the general advantages and disadvantages of hardware before they can capably evaluate it.

Based on the current inventory of computer equipment and adequate estimates of current and forecasted workloads, the next step in the process is to consider the kinds of equipment available that appear to meet projected needs. Information from vendors on possible systems and system configurations becomes more pertinent at this stage and should be reviewed with management and users.

In addition, workloads can be simulated and run on different systems, including those already used in the organization. This process is referred to as *benchmarking*.

Criteria that the systems analysts and users should use to evaluate performance of different systems hardware include the following:

1. The time required for average transactions (including how long it takes to input data and how long it takes to receive output)
2. The total volume capacity of the system (how much can be processed at the same time before a problem arises)
3. The idle time of the CPU or network
4. The size of the memory provided

Some criteria will be shown in formal demonstrations; some cannot be simulated and must be gleaned from manufacturers' or service providers' specifications. It is important to be clear about the required and desired functions before getting too wrapped up in vendors' claims during demonstrations.

Once functional requirements are known and the products currently available are comprehended and compared with what already exists in the organization, the systems analysts in conjunction with users and management can make decisions about whether obtaining new hardware is necessary.

ADVANTAGES OF BUYING COMPUTER HARDWARE If the organization purchases its own computers, it will have full control regarding the type of hardware and software. The company will decide when to purchase it and when to replace it. Often it is cheaper in the long run to buy computer equipment outright, but only if the organization buys the appropriate computer equipment. Buying computers often has tax advantages in the United States because of the depreciation rules.

DISADVANTAGES OF BUYING COMPUTER HARDWARE Purchasing computers presents a problem because the initial cost of buying the equipment is often very high. Phasing in computer equipment over time doesn't always work due to compatibility problems that exist with different models and legacy devices, but buying everything at once means tying up capital or borrowing money. The risk of equipment obsolescence is a serious concern, and getting stuck with equipment that is not useful because someone made the wrong decision is a very large risk. Finally, the organization needs to remember that the full responsibility for the operation and maintenance stays with the company that buys the equipment. This can be a big disadvantage because it ties up people and money.

Renting Time and Space in the Cloud

Instead of purchasing their own equipment, companies can use cloud services. Some of the services available are Web hosting, email hosting, application hosting, backup, storage and processing of databases, archiving, and ecommerce support. The advantages and disadvantages to buying hardware versus using cloud services are shown in Figure 3.6.

Cloud services are scalable, which means if the business grows, the organization can easily add computing power without procuring new hardware. Although few organizations save money

	Advantages	Disadvantages
Buying computer hardware	• Full control over hardware and software • Often cheaper in long run • Provides tax advantages through depreciation	• Initial cost is high • Risk of obsolescence • Risk of being stuck if choice was wrong • Full responsibility for operation and maintenance
Using cloud services	• Maintenance and upgrades performed by provider • Ability to change software and hardware rapidly • Scalable—can grow quickly • Consistent over multiple platforms • No capital is tied up	• Company doesn't control its own data • Data security is at risk • Reliability risks of the Internet platform • Proprietary APIs and software may make switching providers difficult

FIGURE 3.6

Comparing the advantages and disadvantages of buying computer resources versus using cloud services.

by purchasing cloud services, they prize the agility afforded by being able to add or stop cloud services as needed.

Three main categories of cloud computing are Software as a Service (SaaS), Infrastructure as a Service (IaaS), and Platform as a Service (PaaS). With SaaS, the cloud vendor sells access to applications (discussed later in this chapter, when we discuss software). With IaaS, also called Hardware as a Service (HaaS), an organization outsources its hardware operations needs to the service provider. Using the PaaS approach, companies use the Internet to rent all manner of hardware, operating systems, storage, and even network capacity. Another option is to use a hybrid approach in which some of the hardware is maintained in the local organization and some of the software is cloud-based. This includes public and private cloud services.

Decisions on cloud computing can first be addressed on a strategic level. Cearley (2010) states that a business should (1) define a high-level business case that focuses on the high-level benefits of cloud computing for the organization; (2) define core requirements (perhaps making a quick list to start) that support the migration of IT to the cloud, including dimensions such as system performance, security desired in cloud relationships, IT governance, and predicted growth of the enterprise; and (3) define core technologies for the enterprise, starting with a list of cloud computing technologies that are helpful. As an analyst, you may be asked to facilitate any or all of these steps, or your help might be solicited at some point in the process of moving computing services to the cloud.

BENEFITS OF CLOUD COMPUTING Some of the benefits of cloud computing include less time spent maintaining legacy systems or even performing routine tasks such as maintenance or upgrading of present systems. Using the cloud may make it simpler to acquire IT services and may also make it easier and quicker to separate from or discontinue services that are no longer necessary. Using cloud services makes your applications scalable, which means you can grow them easily by adding more cloud resources. Cloud computing also has the potential to offer consistency across multiple platforms that were previously disjointed or difficult to integrate. Finally, no capital is tied up, and no financing is required.

DRAWBACKS OF CLOUD COMPUTING As with any other innovation, there are some disadvantages of adopting cloud computing for organizations. Loss of control of data stored in the cloud is perhaps the most striking concern when relying on cloud computing; if the cloud services provider ceased to exist, it is unclear what would happen to the organization's data. Looming almost as large are potential security threats to data that is not stored on premises or even on the organization's own computers. Reliability of the Internet as a platform forms the third main concern for organizations.

Furthermore, something that could dampen an organization's maneuverability when it comes to cloud computing is the prospect that the company could get stuck using application

programming interfaces (APIs) that are proprietary with the cloud provider. This might make switching among cloud computing providers more difficult. Another development in cloud computing of which analysts should be aware is that some countries are now mandating that any company that does business there must use cloud service providers based in that country. For example, Apple is now obeying legislation like this in China.

Evaluation of Vendor Support for Computer Hardware

Several key areas need to be evaluated when weighing the support services vendors offer to businesses. Most hardware vendors offer testing of hardware on delivery and a 90-day warranty covering any factory defects, but you must ascertain what else the vendor has to offer. Cloud service providers offer 30- or 60-day money-back guarantees. Vendors of comparable quality frequently distinguish themselves from others in the range of support services they offer.

A list of key criteria that ought to be checked when evaluating vendor support is provided in Figure 3.7. Support services include routine and preventive maintenance of hardware, specified response time (within six hours, next working day, and so on) in case of emergency equipment breakdowns, loan of equipment in the event that hardware must be permanently replaced or off-site repair is required, and in-house training or off-site group seminars for users. Cloud providers make claims for uptime and promises to get back to you on a question within a certain amount of time, such as 24 hours. Peruse the support services documents accompanying the purchase of equipment and remember to involve appropriate legal staff before signing contracts for equipment or services.

Unfortunately, the process of evaluating computer hardware and cloud providers is not as straightforward as simply comparing costs and choosing the least expensive option. Before becoming convinced that buying cheaper compatibles is the way to endow your system with add-on capability, you need to do enough research to feel confident that a hardware vendor or cloud provider is a stable corporate entity.

Understanding the Bring Your Own Device (BYOD) Option

Bring your own device (BYOD) or bring your own technology (BYOT) is a trend appearing in a variety of organizations in different industries. These organizations range in size and have employees of differing skills. In general, BYOD or BYOT means that rather than using a smartphone or tablet issued by the organization, the employee uses his or her own device to access corporate networks, data, and services remotely when working outside the office. The BYOD approach is often presented

FIGURE 3.7

Guidelines for vendor selection.

Vendor Services	Specifics Vendors Typically Offer
Hardware Support	Full line of hardware Quality products Warranty
Software Support	Complete software needs Custom programming Warranty
Installation and Training	Commitment to schedule In-house training Technical assistance
Maintenance	Routine maintenance procedures Specified response time in emergencies Equipment loan while repair is being done
Cloud Services	Web hosting Email hosting Data storage
Software as a Service	Automatic software upgrades Support services Security and antivirus protection

as a way to keep the organization's hardware costs down and improve morale among employees who, on average, already bring one or more mobile devices to work for personal tasks.

As a systems analyst, you will work to discover through a combination of interviews, surveys, and observation what personal technology is being used by the majority of the group. For example, if you observe that all the executives in the decision-making group already bring in iPads every day, you might conclude that designing dashboards for the iPad has some merit. They are devices you want to include in your original design rather than as an afterthought. If your observations show that most executives prefer face-to-face meetings, but they use their handheld devices (such as mobile phones) to check other corporate data and emails during meetings, then you might want to consider supporting corporate email on their personal phones.

BENEFITS AND DRAWBACKS OF BYOD The benefits of BYOD computing include building employee morale, a potential for lowering the initial cost of organization IT hardware purchase, facilitating remote, round-the-clock access to corporate computer networks regardless of location, and building on a familiar user interface to access corporate computing services, applications, databases, and storage. Security risks posed by untrained users are probably the biggest drawback. Identified threats are loss of the device itself, theft of the device and its data, unauthorized access to corporate networks using personal mobile devices, and so on. Other threats are posed by commonly occurring behavior with mobile devices that may be fine on the private side but not on the corporate side, such as using free Wi-Fi hotspots, using apps such as Dropbox, and others.

Creating Custom Software

Analysts and organizations are increasingly faced with a make, buy, or outsource decision when assessing software for information systems projects, particularly when contemplating upgrades to existing or legacy systems. Analysts have three choices: creating their own software, purchasing commercial off-the-shelf (COTS) software, or using software from a SaaS provider. Figure 3.8 summarizes the advantages and disadvantages of each of these options.

	Advantages	Disadvantages
Creating Custom Software	• Specific response to specialized business needs • Innovation may give firm a competitive advantage • In-house staff available to maintain software • Pride of ownership	• May be significantly higher initial cost compared to COTS software or ASP • Necessity of hiring or working with a development team • Ongoing maintenance
Purchasing COTS Packages	• Refined in the commercial world • Increased reliability • Increased functionality • Often lower initial cost • Already in use by other firms • Help and training comes with software	• Programming focused; not business focused • Must live with the existing features • Limited customization • Uncertain financial future of vendor • Less ownership and commitment
Using SaaS	• Organizations that do not specialize in information systems can focus on what they do best (their strategic mission) • There is no need to hire, train, or retain a large IT staff • There is no expenditure of employee time on nonessential IT tasks	• Loss of control of data, systems, IT employees, and schedules • Concern over the financial viability and long-run stability of the SaaS provider • Security, confidentiality, and privacy concerns • Loss of potential strategic corporate advantage regarding innovativeness of applications

FIGURE 3.8

Comparing the advantages and disadvantages of creating custom software, purchasing COTS packages, and outsourcing to a SaaS provider.



CONSULTING OPPORTUNITY 3.2

Veni, Vidi, Vendi, or “I Came, I Saw, I Sold”

“It’s really some choice. I mean, no single package seems to have everything we want. Some of them come darn close, though,” says Roman, an advertising executive for *Empire Magazine* with whom you have been working on a systems project. Recently, the two of you have decided that packaged software would probably suit the advertising department’s needs and stem its general decline.

“The last guy’s demo we saw—you know, the one who worked for Data Coliseum—really had a well-rounded pitch. And I like their brochure. Full-color printing, on card stock. Classic,” Roman asserts. “And what about those people from Vesta Systems? They’re really fired up. And their package was easy to use with a minimum of ceremony. Besides, they said they would train all 12 of us, on-site, at no charge. But look at their advertising. They just take things off their printers.”

Roman fiddles in his chair as he continues his ad hoc review of software and software vendors. “That one package from Mars, Inc. really sold me all on its own, though. I mean, it had a built-in

calendar widget to track ad campaign roll outs. And I like the way the menus for the screen displays could all be chosen by Roman numerals. It was easy to follow. And the vendor isn’t going to be hard to move on price. I think they’re already in a price war.”

“Do you want to know my favorite, though?” Roman asks archly. “It’s the one put out by Jupiter, Unlimited. I mean, it has everything, doesn’t it? It costs a little extra coin, but it does what we need it to do, and the documentation is heavenly. They don’t do any training, of course. They think they’re above it.”

To answer Roman’s burning questions by your March 15 deadline, you need to evaluate the software as well as the vendors, systematically, and then render a decision. Evaluate each vendor and package based on what Roman has said so far. (Assume that you can trust his opinions.) What are Roman’s apparent biases when evaluating software and vendors? What further information do you need about each company and its software before you can make a selection? Set up a table to evaluate each vendor. Answer each question in a separate paragraph.

Several situations call for the creation of original, custom software or software components. The most likely instance is when COTS software does not exist or cannot be identified for the desired application. Alternatively, the software may exist but be unaffordable or not easily purchased or licensed.

Original software should be created when an organization is attempting to gain a competitive advantage through the leveraged use of information systems. This is often the case when an organization is creating ecommerce or other innovative applications where none have existed. An organization also may be a “first mover” in the use of a particular technology or in its particular industry. Organizations that have highly specialized requirements or exist in niche industries also may benefit from creating original software.

The advantages of creating custom software include being able to respond to specialized user and business needs, gaining a competitive advantage by creating innovative software, having in-house staff available to maintain the software, and pride of owning something you have created.

The drawbacks of developing custom software include the potential for a significantly higher initial cost compared to purchasing COTS software or contracting with a SaaS provider, the necessity of hiring or working with a development team, and the fact that you are responsible for the ongoing maintenance because you created the software.

Purchasing COTS Software

COTS software includes such products as the Microsoft Office suite, which includes Word for word processing, Excel for spreadsheets, Access for building databases, and other applications. Other types of COTS software are for organization-level systems rather than office or personal use. A website such as softwareadvice.com lists popular (but costly) enterprise resource planning (ERP) packages such as Oracle and SAP in their reviews of COTS software. These packages differ radically in the amount of customization, support, and maintenance required compared to Microsoft Office. COTS software also can refer to software components or objects (also called building blocks) that can be purchased to provide a particular needed functionality in a system.

You should consider using COTS software when you can easily integrate the applications or packages into existing or planned systems and when you have identified no need to immediately or continuously change or customize systems for users. Your forecasts should demonstrate that the organization you are designing the system for is unlikely to undergo major changes after the

proposed purchase of COTS software, such as a dramatic increase in customers or large physical expansions.

There are some advantages to purchasing COTS software that you should keep in mind as you weigh alternatives. One advantage is that these products have been refined through the process of commercial use and distribution and additional functionalities often are offered. Another advantage is that packaged software is typically extensively tested and thus extremely reliable.

Increased functionality is often offered with COTS software because a commercial product is likely to have companion products, add-on features, and upgrades that enhance its attractiveness. Also, analysts often find that the initial cost of COTS software is lower than the cost for either in-house software development or the use of a SaaS provider.

Another advantage of purchasing COTS packages is that many other companies use them, so analysts are not experimenting on their clients with one-of-a-kind software applications. Finally, COTS software boasts an advantage in the help and training that accompanies the purchase of the packaged software.

One example of the use of COTS software is from a theatre company in the nonprofit sector, in which organizations (particularly in the performing arts) tend to lag behind their for-profit counterparts in adoption of information communication technologies (ICTs). The theatre company was predictably slow to move to the Web. When the company wanted to create ecommerce applications, it was put in a position of having to hire outside designers to create ecommerce applications for them. In light of the expense and lack of in-house expertise, many nonprofit organizations simply did not move the business portion of their organizations to the Web; instead, they waited for COTS packages, such as PC-based box office software, or SaaS, such as online ticketing agencies with automation already in place, to make these services available to patrons. In-house, custom software development is out of the question for most of these groups, which typically have small or nonexistent IT staffs and budgets and minimal internal IT expertise.

There is a downside to the use of COTS software. Because it is not meant to be fully customizable, the theatre company lost its ability to change the software to include key features in its donor database on which users relied. COTS software also may include errors that could expose an organization to liability issues.

There are other disadvantages to consider with the purchase of COTS software, including the fact that packages are programmed rather than being focused on human users working in a business. Users must live with whatever features exist in the software, whether they are appropriate or not. A disadvantage that grows out of this is the limited customizability of most packaged software. Other disadvantages to purchasing COTS software include the necessity of investigating the financial stability of the software vendor and the diminished sense of ownership and commitment that is inevitable when the software is considered a product rather than a process.

To achieve some perspective on systems being developed, you should recognize that over half of all projects are built from the ground up (two-thirds using traditional methods like SDLC and prototyping and one-third using agile or object-oriented technologies). Most of these are developed using an internal systems analysis team. Computer programmers may be in-house or outsourced.

Fewer than half of all projects are developed from existing applications or components. The great majority are modified, some extensively. Less than 5 percent of software is off-the-shelf software that requires no modifications at all.

Using the Services of a SaaS Provider

Organizations may realize some benefits from taking an entirely different approach to procuring software: Outsourcing some of the organization's software needs to a SaaS provider that specializes in IT applications.

There are specific benefits to outsourcing applications to a SaaS provider. For example, organizations that want to retain their strategic focus and do what they're best at may want to outsource the production of information systems applications. In addition, outsourcing software needs means that the organization doing the outsourcing may be able to sidestep the need to hire, train, and retain a large IT staff. This can result in significant savings. When an organization uses SaaS, there is little or no expenditure of valuable employee time on nonessential IT tasks; these are handled professionally by SaaS.

Hiring a SaaS provider should not be considered a magic formula for addressing software requirements. Some drawbacks to the use of SaaS must be seriously considered. One disadvantage is a general loss of control over corporate data, information systems, IT employees, and even processing and project schedules. Some companies believe that the heart of their business is their information, so even the thought of relinquishing control over it is distressing. Another disadvantage is concern over the financial viability of any SaaS that is chosen. There might also be concerns about the security of the organization's data and records, along with concern about confidentiality of data and client privacy. Finally, when choosing a SaaS provider, there is a potential loss of strategic corporate advantage that might have been gained through the company's deployment of its own innovative applications created by its employees.

Evaluation of Vendor Support for Software and SaaS

Whether you purchase a COTS package or contract for SaaS from a provider, you will be working with vendors who may have their own best interests at heart. You must be willing to evaluate software with users and not be unduly influenced by vendors' sales pitches. Specifically, there are six main categories on which to grade software, as shown in Figure 3.9: performance effectiveness, performance efficiency, ease of use, flexibility, quality of documentation, and manufacturer support.

Evaluate packaged software based on a demonstration using test data from your client's business and examine accompanying documentation. Vendors' descriptions alone will not suffice. Vendors typically certify that software is working, but they do not guarantee that it will be error-free in every instance, that it will not crash when users take incorrect actions, or that it will be compatible with all other software the organization is currently running. Obviously, they will not guarantee their packaged software if it is used in conjunction with faulty hardware.

FIGURE 3.9

Guidelines for evaluating software.

Software Requirements	Specific Software Features
Performance Effectiveness	Able to perform all required tasks Able to perform all tasks desired Well-designed display screens Adequate capacity
Performance Efficiency	Fast response time Efficient input Efficient output Efficient storage of data Efficient backup
Ease of Use	Satisfactory user interface Help menus available "Read Me" files for last-minute changes Flexible interface Adequate feedback Good error recovery
Flexibility	Options for input Options for output Usable with other software
Quality of Documentation	Good organization Adequate online tutorial Website with FAQ
Manufacturer Support	Technical support hotline Newsletter/email Website with downloadable product updates

Identifying, Forecasting, and Comparing Costs and Benefits

Costs and benefits of a proposed computer system must always be considered together because they are interrelated and often interdependent. Although a systems analyst tries to propose a system that fulfills various information requirements, decisions to continue with the proposed system will be based on a cost-benefit analysis, not on information requirements. In many ways, benefits are measured by costs, which becomes apparent in the next section.

Forecasting

Systems analysts are required to predict certain key variables before submitting a proposal to the client. To some degree, a systems analyst will rely on a what-if analysis such as, “What if labor costs rise only 5 percent per year for the next three years, rather than 10 percent?” The systems analyst should realize, however, that he or she cannot rely on what-if analysis for everything if the proposal is to be credible, meaningful, and valuable.

A systems analyst has many forecasting models available. The main condition for choosing a model is the availability of historical data. If historical data are unavailable, the analyst must turn to one of the judgment methods: estimates from the sales force, surveys to estimate customer demand, Delphi studies (a consensus forecast developed independently by a group of experts through a series of iterations), creation of scenarios, or historical analogies.

If historical data are available, the next differentiation between classes of techniques involves whether the forecast is conditional or unconditional. Conditional implies that there is an association among variables in the model or that such a causal relationship exists. Common methods in this group include correlation, regression, leading indicators, econometrics, and input/output models.

Unconditional forecasting means the analyst isn’t required to find or identify any causal relationships. Consequently, systems analysts find that these methods are low-cost, easy-to-implement alternatives. Included in this group are graphical judgment, moving averages, and analysis of time-series data. Because these methods are simple, reliable, and cost-effective, the remainder of the section focuses on them.

ESTIMATION OF TRENDS Trends can be estimated in a number of different ways. One way to estimate trends is to use a moving average. This method is useful because some seasonal, cyclical, or random patterns may be smoothed, leaving the trend pattern. The principle behind moving averages is to calculate the arithmetic mean of data from a fixed number of periods; a three-month moving average is simply the average of the past three months. For example, the average sales for January, February, and March are used to predict the sales for April. Then the average sales for February, March, and April are used to predict the sales for May, and so on.

When the results are graphed, it is easily noticeable that the widely fluctuating data are smoothed. The moving average method is useful for its smoothing ability, but it also has many disadvantages. Moving averages are more strongly affected by extreme values than by using graphical judgment or estimating by using other methods, such as least squares. An analyst should learn forecasting well, as it often provides information that is valuable in justifying an entire project.

Identifying Benefits and Costs

Benefits and costs can be either tangible or intangible. Both tangible and intangible benefits and costs must be taken into account when systems are considered.

TANGIBLE BENEFITS Tangible benefits are advantages that are measurable in dollars that accrue to the organization through the use of the information system. Examples of tangible benefits are an increase in the speed of processing, access to otherwise inaccessible information, access to information on a more timely basis than was possible before, the advantage of the computer’s superior calculating power, and a decrease in the amount of employee time needed to complete specific tasks. And there are other tangible benefits. Although measurement is not always easy, tangible benefits can be measured in terms of actual dollars, resources, or time saved.



CONSULTING OPPORTUNITY 3.3

We're Off to See the Wizards

Elphaba I. Menzel and Glinda K. Chenoweth are the owners of Emerald City Beautyscapes, a commercial landscaping company. They are trying to decide whether to write their own software, perhaps using Microsoft Access as a basis; adopt a COTS software package such as QuickBooks Pro; or hire a service called Lawn Wizards, Inc. to perform all their bookkeeping functions.

Elphaba turns to Glinda and asks, "Is it possible for us to create a system of our own?"

Glinda replies, "I suppose we could, but it would take forever. We would need to define all our fields, our queries, and our reports. We would need to know who hasn't paid us yet and how long it has been since we last billed them."

"Yes," says Elphaba, "and we would also have to create product descriptions, service descriptions, and codes for everything we sell and provide."

"If that were all we needed, we could probably do it," says Glinda. "But we also need to include a scheduling system. We need to know when we can provide the services to our customers and what to do if we fall behind schedule. Maybe it just isn't worth it."

"Still," reflects Glinda, "my mother used to say 'There's no place like home.' Maybe there's no software like homegrown."

"You see both sides of everything," remarks Elphaba. "But the path you want to take is too long and risky. We need a software package that is ready for us to use now. I hear that there are products they call commercial off-the-shelf (COTS) software that we can buy and adapt to our lawn service business. I'll investigate." So Elphaba sets out to look for software that may be suitable.

"I've found something," cries Elphaba. "I found this software called QuickBooks Pro at www.quickbooks.intuit.com, and it looks like we can afford it. There are numerous versions of the software already—one for accounting, one for construction, one for health services. Maybe we can find a package that suits us. If not, it looks like we can customize the generic version of QuickBooks Pro to fit our needs."

"Our system could grow too. QuickBooks Pro is readily scalable. We can add customers, suppliers, or products easily. I just wanted to plant the idea of buying a ready-made package on you. Another possibility is to subscribe to a QuickBooks Online plan."

"That's interesting," says Glinda, "but I've been doing my own research. Some of our competitors have told me they let a company do all the work for them. The company is called Lawn Wizards. They do landscaping, but they also maintain accounts receivable and scheduling packages."

So off they went to see the Wizards.

Joel Green, the owner and creator of Lawn Wizards, is proud of his software. "I spent a great deal of time working with my suppliers, that is, nurseries, in the area, and we have developed a coding system for everything," he brags. "All the trees, sizes of trees, shrubs, flowers, mulch, and even lawn care tools have numbers."

"I started with a small firm, but when customers realized I paid attention to every little detail, my business blossomed." He adds, "My suppliers love my system because it cuts down on confusion."

"I noticed that my competitors were working with the same suppliers but were getting less preferential treatment because they couldn't communicate about products very effectively. So I decided I would offer my software for hire. I would make money by renting out my software and demand even greater respect from my suppliers. I can even deliver it through the cloud. My end-user license agreement states that I own the software, product codes, and data generated by the system."

"Using my unique Wizards software, I can customize the package a bit for the customer, but essentially all the lawn services in the state will be using my database, codes, and B2B features. I maintain my software. If you could see the software code, it would look just like a manicured lawn."

Now Glinda and Elphaba are even more confused than before. They have four distinct options: create a custom package on their own, buy a standalone COTS software product such as QuickBooks Pro, subscribe to an online plan such as QuickBooks Online, or outsource their needs to Lawn Wizards. Help them learn the true secret of (software) happiness by helping them articulate the pros and cons of each of their alternatives. What would you recommend? In two paragraphs, write a recommendation that grows out of your consideration of their specific business situation.

INTANGIBLE BENEFITS Some benefits that accrue to an organization from the use of an information system are difficult to measure but are important nonetheless. They are known as intangible benefits.

Intangible benefits include improving the decision-making process, enhancing accuracy, becoming more competitive in customer service, maintaining a good business image, and increasing job satisfaction for employees by eliminating tedious tasks. As you can see from this list, intangible benefits are extremely important and can have far-reaching implications for a business as it relates to people both outside and within the organization.

Although intangible benefits of an information system are important factors that must be considered when deciding whether to proceed with a system, a system built solely for its intangible benefits will not be successful. You must discuss both tangible and intangible benefits in your proposal because presenting both will enable decision makers in the business to make a well-informed decision about the proposed system.

TANGIBLE COSTS The concepts of tangible and intangible costs present a conceptual parallel to the tangible and intangible benefits discussed already. Tangible costs are costs that a systems analyst and the business's accounting personnel can accurately project.

Included in tangible costs are the cost of equipment such as computers and printers, the cost of resources, the cost of systems analysts' time, the cost of computer programmers' time, and other employees' salaries. These costs are usually well established or can be discovered quite easily; they are costs that will require the business to make a cash outlay.

INTANGIBLE COSTS Intangible costs are difficult to estimate and may not be known. They include losing a competitive edge, losing the reputation for being first with an innovation or the leader in a field, declining company image due to increased customer dissatisfaction, and ineffective decision making due to untimely or inaccessible information. As you can imagine, it is nearly impossible to accurately project a dollar amount for intangible costs. To aid decision makers who want to weigh a proposed system and all its implications, you must include intangible costs even though they are not quantifiable.

Comparing Costs and Benefits

There are many well-known techniques for comparing the costs and benefits of a proposed system. Two are break-even analysis and payback.

BREAK-EVEN ANALYSIS By comparing costs alone, a systems analyst can use break-even analysis to determine the break-even capacity of a proposed information system. The point at which the total costs of the current system and the proposed system intersect represents the break-even point, the point where it becomes profitable for the business to get the new information system.

Total costs include the costs that recur during operation of a system plus the developmental costs that occur only once (one-time costs of installing a new system)—that is, the tangible costs that were discussed previously. Figure 3.10 illustrates a break-even analysis on a small store that maintains inventory using a manual system. As volume rises, the costs of the manual system rise at an increasing rate. A new computer system would cost a substantial sum up front, but the incremental costs for higher volume would be rather small. The graph shows that the computer system would be cost-effective if the business sold about 600 units per week.

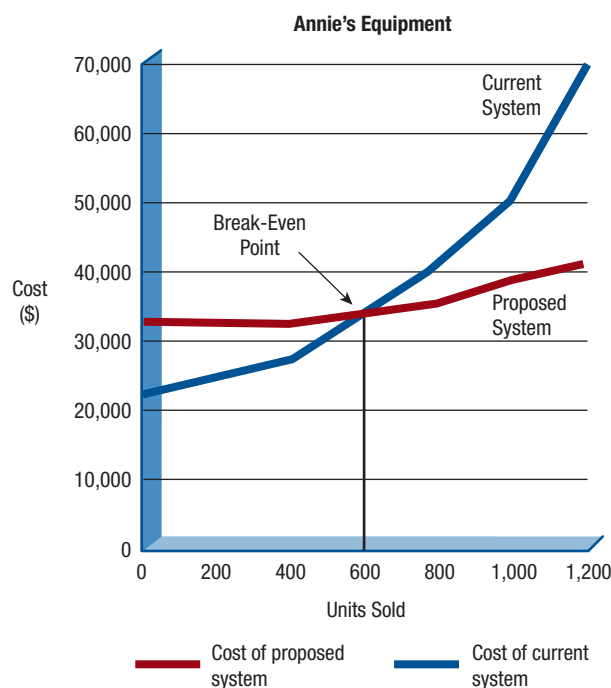
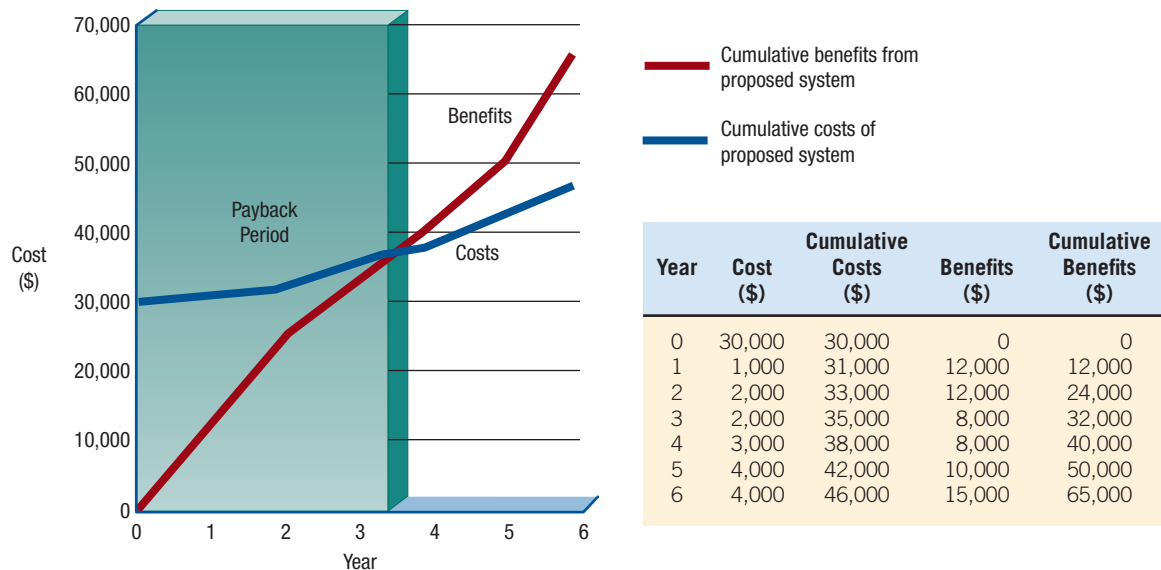


FIGURE 3.10

Break-even analysis for a proposed inventory system.

**FIGURE 3.11**

Payback analysis showing a payback period of three and a half years.

Payback is useful when a business is growing and volume is a key variable in costs. One disadvantage of break-even analysis is that benefits are assumed to remain the same, regardless of which system is in place. From our study of tangible and intangible benefits, we know that this is clearly not the case.

Payback analysis can determine how long it will take for the benefits of the system to pay back the costs of developing it. Figure 3.11 illustrates a system with a payback period of three and a half years.

Managing Time and Activities

The process of analysis and design can become unwieldy, especially when the system being developed is large. To keep the development activities as manageable as possible, systems analysts usually use some of the techniques of project management to help get organized.

The Work Breakdown Structure

Systems analysts are responsible for completing projects on time and within budget and for including the features promised. To accomplish all three of these goals, a project needs to be broken down into smaller tasks or activities. These tasks together make up a work breakdown structure (WBS).

When defined properly, the tasks that comprise a WBS have special properties:

1. Each task or activity contains one deliverable, or tangible outcome, from the activity.
2. Each task can be assigned to a single individual or a single group.
3. Each task has a responsible person monitoring and controlling performance.

Activities in a work breakdown structure do not need to take the same amount of time or involve the same number of team members. The activities defined must add up to 100 percent of the work, however.

The main method for developing a WBS is decomposition, or starting with large ideas and then breaking them down into manageable activities. This subdivision of ideas into smaller ideas and eventually into tasks stops when each task has only one deliverable.

There are different types of work breakdown structures. A WBS can be product oriented. In other words, building a website can be broken down into many parts, with each set of pages having a specific purpose. You could divide a website into its home page, product description pages, a FAQ page, a contact page, and an ecommerce page. Each of these pages could contain activities that you could use in your work breakdown structure.

1.0 Project Initiation
1.1 Assemble and welcome project team
1.2 Conduct historical research about the business
1.3 Discuss objectives with client
2.0 Early Planning Phase
2.1 Investigate feasibility
2.2 Consider a make vs. buy decision
2.3 Develop a work breakdown structure
2.4 Provide time estimates
2.5 Develop a project schedule
2.6 Calculate cost estimates
2.7 Prepare project proposal to client
2.8 Present proposal to client
3.0 Develop supporting plans
3.1 Develop a quality management plan
3.2 Identify risks and build a risk management plan
3.3 Describe a communications plan
3.4 Develop a procurement plan
4.0 Analysis
4.1 Conduct interviews of key personnel
4.2 Administer questionnaires
4.3 Read company reports
4.4 Analyze data flow
5.0 Design
5.1 Build prototype website
5.2 Obtain reactions from client
5.3 Modify prototype website
5.4 Seek final recommendations from client
5.5 Complete website
6.0 Launch
6.1 Create training manual
6.2 Document website features and logic
6.3 Present final website to client

FIGURE 3.12

A sample work breakdown structure (WBS).

Another way is to create a process-oriented work breakdown structure. An example of this is shown in Figure 3.12. This type of WBS is typical in systems analysis and design. In this example, we show the development of a website, but rather than show the development of each page, this example emphasizes the importance of each phase in the systems development life cycle.

Time Estimation Techniques

The process of analysis and design can become unwieldy, especially when the system being developed is large. To keep the development activities as manageable as possible, you can employ some of the techniques of project management to help get organized. In this section, we discuss keeping a project on schedule by using time management techniques and keeping a project on budget by using cost management and control techniques.

One of the difficult tasks is estimating the time it takes to complete each of the tasks. There are numerous approaches:

1. Relying on experience
2. Using analogies
3. Using three-point estimation
4. Identifying function points
5. Using time estimation software

RELYING ON EXPERIENCE Experience pays off when it comes to estimating how long an activity will take. If you have previous experience developing software, you will not only know how much time some tasks will likely take, you will also know how much time it will take if something goes wrong. Your experience gives you a most likely estimate as well as a pessimistic estimate.

USING ANALOGIES If you don't have experience developing a particular piece of software but have worked on other types of projects of any kind, you still may be able to arrive at meaningful estimates. This approach involves identifying a project that is in some ways similar to the one you are about to begin and then describing the analogy. This means you will need to build two models, including two Program Evaluation and Review Technique (PERT) or network diagrams, and compare their similarities. Then you can have confidence when you provide time estimates for the new project.

USING THREE-POINT ESTIMATION Three-point estimation has been used for many years and is still a valid technique for time estimating. You start by developing three time estimates for completing each task and then apply a simple formula to calculate a weighted average. The formula is:

$$E = (a + 4 * m + b)/6$$

where a is the best-case estimate, b is the pessimistic or worst-case time estimate, and m is the most likely time estimate.

Typically the best-case scenario would improve on the time elapsed a little, but the pessimistic or worst-case estimate would imply that something disastrous happens, such as a weather or personnel delay. For example, writing a software module may take 10 days in most cases, but if the project is easier than expected, it might take only 8 days. If, however, the coder accepts a job elsewhere, it may take 30 days. If you apply the formula, you get this estimate:

$$\begin{aligned} E &= (8 + 4 * 10 + 30)/6 \\ &= 13 \text{ days} \end{aligned}$$

In this case, you will estimate that it will take 13 days to finish the task.

USING FUNCTION POINT ANALYSIS Another way of estimating the amount of work that needs to be done and how large a staff you need to complete a project is called function point analysis. This method takes the five main components of a computer system—(1) external inputs, (2) external outputs, (3) external queries, (4) internal logical files, and (5) external interface files—and rates them in terms of complexity.

You can use function point analysis to estimate the time it takes to develop a system in different computer languages and compare them to one another. For more information about function point analysis, visit the International Function Point Users Group's website at www.ifpug.org.

USING TIME ESTIMATION SOFTWARE Estimating models, such as the Constructive Cost Model (COCOMO II), Constructive Systems Engineering Cost Model (COSYSMO), or software based on one or both of these models, such as SystemStar, work as follows. A systems analyst enters an estimate of the size of the system. This can be entered in a number of different ways, including the lines of source code of the current system.

Also considered are other variables, such as the experience or capability of the team, the type of platform or operating system, the level of usability of the finished software (for example, what languages are necessary), and other factors that can drive up costs. Once the data are entered, you can make calculations and get a rough projection of the completion date. As the project gets under way, more specific estimates are possible.



CONSULTING OPPORTUNITY 3.4

Food for Thought

“We could really make some changes. Shake up some people. Let them know we’re with it. Technologically, I mean,” said Malcolm Warner, vice president for AllFine Foods, a wholesale dairy products distributor. “That old system should be overhauled. I think we should just tell the staff that it’s time to change.”

“Yes, but what would we actually be improving?” Kim Han, assistant to the vice president, asks. “I mean, there aren’t any substantial problems with the system input or output that I can see.”

Malcolm snaps, “Kim, you’re purposely not seeing my point. People out there see us as a stodgy firm. A new computer system could help change that. Change the look of our invoices. Send jazzier reports to the food store owners. Get some people excited about us as leaders in wholesale food distributing and computers.”

“Well, from what I’ve seen over the years,” Kim replies evenly, “a new system is very disruptive, even when the business really needs it. People dislike change, and if the system is performing the way it should, maybe there are other things we could do to update our image that wouldn’t drive everyone nuts in the process. Besides, you’re talking big bucks for a new gimmick.”

Malcolm says, “I don’t think just tossing it around here between the two of us is going to solve anything. Check on it and get back to me. Wouldn’t it be wonderful?”

A week later, Kim enters Malcolm’s office with several pages of interview notes in hand. “I’ve talked with most of the people who have extensive contact with the system. They’re happy, Malcolm. And they’re not just talking through their hats. They know what they’re doing.”

“I’m sure the managers would like to have a newer system than the guys at Quality Foods,” Malcolm replies. “Did you talk to them?”

Kim says, “Yes. They’re satisfied.”

“And how about the people in systems? Did they say the technology to update our system is out there?” Malcolm inquires insistently.

“Yes. It can be done. That doesn’t mean it *should* be,” Kim says firmly.

As the systems analyst for AllFine Foods, how would you assess the feasibility of the systems project Malcolm is proposing? Based on what Kim has said about the managers, users, and systems people, what seems to be the operational feasibility of the proposed project? What about the economic feasibility? What about the technological feasibility? Based on what Kim and Malcolm have discussed, would you recommend that a full-blown systems study be done? Answer yes or no, and then discuss your recommendation and the reasoning behind it in a paragraph.

Project Scheduling

Planning includes all the activities required to select a systems analysis team, assign members of the team to appropriate projects, estimate the time required to complete each task, and schedule the project so that tasks are completed in a timely fashion. Control means using feedback to monitor the project, including comparing the plan for the project with its actual evolution. In addition, control means taking appropriate action to expedite or reschedule activities to finish on time while motivating team members to complete the job properly.

This section contains an example in which a systems analyst, acting as a project manager, begins with the basic activities of analysis, design, and implementation. Then the analyst uses decomposition to break apart the main activities into smaller subtasks, as shown in Figure 3.13. Then the analysis phase is further broken down into data gathering, data flow and decision

Phase	Activity
Analysis	Data gathering Data flow and decision analysis Proposal preparation
Design	Data entry design Input design Output design Data organization
Implementation	Implementation Evaluation

Break apart
the major
activities into
smaller ones.

FIGURE 3.13

Beginning to plan a project by breaking it into three major activities.

FIGURE 3.14

Refining the planning and scheduling of analysis activities by adding detailed tasks and establishing the time required to complete the tasks.

Activity	Detailed Activity	Weeks Required
Data gathering	Conduct interviews	3
	Administer questionnaires	4
	Read company reports	4
	Introduce prototype	5
	Observe reactions to prototype	3
Data flow and decision analysis	Analyze data flow	8
Proposal preparation	Perform cost-benefit analysis	3
	Prepare proposal	2
	Present proposal	2

analysis, and proposal preparation. Design is broken down into data entry design, input and output design, and data organization. The implementation phase is divided into implementation and evaluation.

In subsequent steps, a systems analyst needs to consider each of these tasks and break them down further so that planning and scheduling can take place. Figure 3.14 shows how the analysis phase is described in more detail. For example, data gathering is broken down into five activities, from conducting interviews to observing reactions to the prototype. This particular project requires data flow analysis but not decision analysis, so the systems analyst has written in “analyze data flow” as the single step in the middle phase. Finally, proposal preparation is broken down into three steps: perform cost-benefit analysis; prepare proposal, and present proposal.

The systems analyst, of course, has the option to break down steps further. For instance, the analyst could specify each of the people to be interviewed. The amount of detail necessary depends on the project, but all critical steps need to appear in the plans.

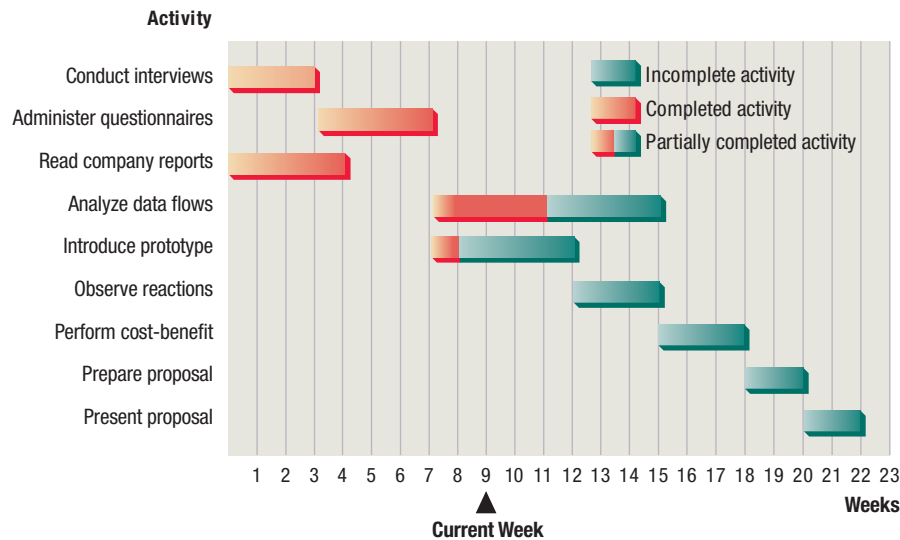
Sometimes the most difficult part of project planning is the crucial step of estimating the time it takes to complete each task or activity. When quizzed about reasons for lateness on a particular project, project team members cited poor scheduling estimates that hampered the success of projects from the outset. There is no substitute for experience in estimating time requirements, and systems analysts who have had the opportunity of an apprenticeship are fortunate in this regard.

Planners have attempted to reduce the inherent uncertainty in determining time estimates by projecting most likely, pessimistic, and optimistic estimates and then using a weighted average formula to determine the expected time an activity will take. This approach offers little more in the way of confidence, however. Perhaps the best strategy for a systems analyst is to adhere to a structured approach in identifying activities and describing these activities in sufficient detail. In this manner, the systems analyst will at least be able to limit unpleasant surprises.

Using Gantt Charts for Project Scheduling

A Gantt chart is a tool that enables you to easily schedule tasks. It is a chart on which bars represent tasks or activities. The length of each bar represents the relative length of the task.

Figure 3.15 is an example of a two-dimensional Gantt chart in which time is indicated on the horizontal dimension, and a description of activities makes up the vertical dimension. In this example, the Gantt chart shows the analysis or information-gathering phase of the project. Notice on the Gantt chart that conducting interviews will take three weeks, administering the questionnaire will take four weeks, and so on. These activities overlap part of the time. In the

**FIGURE 3.15**

Using a two-dimensional Gantt chart for planning activities that can be accomplished in parallel.

chart the special symbol ▲ signifies that right now, it is week 9. The bars with color shading represent projects or parts of projects that have been completed, telling us that the systems analyst is behind in introducing prototypes but ahead in analyzing data flows. Action must be taken on introducing prototypes soon so that other activities or even the project itself will not be delayed as a result.

The main advantage of a Gantt chart is its simplicity. Not only is this technique easy to use, but it also lends itself to worthwhile communication with end users. Another advantage of using a Gantt chart is that the bars representing activities or tasks are drawn to scale; that is, the size of the bars indicates the relative length of time it will take to complete each task. You can use Microsoft Project to develop a Gantt chart.

Using PERT Diagrams

PERT is an acronym for Program Evaluation and Review Technique. A program (a synonym for a project) is represented by a network of nodes and arrows that are evaluated to determine the critical activities, improve the schedule if necessary, and review progress once the project is undertaken. PERT was developed in the late 1950s for use in the U.S. Navy's *Polaris* nuclear submarine project. It reportedly saved the U.S. Navy two years of development time. PERT charts are called network diagrams in Microsoft Project.

PERT is useful when activities can be done in parallel rather than in sequence. A systems analyst can benefit from using PERT by applying it to systems projects on a smaller scale, especially when some team members can be working on certain activities at the same time that other team members are working on other tasks.

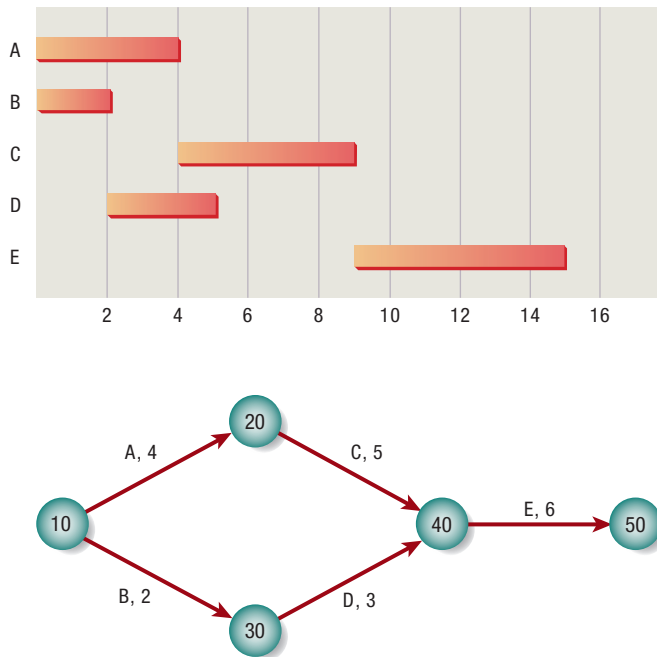
Figure 3.16 compares a simple Gantt chart with a PERT diagram. Arrows in the PERT diagram represent the activities expressed as bars in the Gantt chart. The length of the arrows has no direct relationship with the activity durations. Circles on the PERT diagram are called events and can be identified by numbers, letters, or any other arbitrary form of designation. The circular nodes are present to (1) recognize that an activity is completed and (2) indicate which activities need to be completed before new activities may be undertaken (which is called *precedence*).

In the PERT example shown here, activity C may not be started until activity A is completed. Precedence is not indicated at all in the Gantt chart, so it is not possible to tell whether activity C is scheduled to start on day 4 on purpose or by coincidence.

A project has a beginning, a middle, and an end; in this example, the beginning is event 10, and the end is event 50. To find the length of the project, each path from beginning to end is identified, and the length of each path is calculated. In this example, path 10–20–40–50 has a length of 15 days, whereas path 10–30–40–50 has a length of 11 days. Even though one person may be working on path 10–20–40–50 and another on path 10–30–40–50, the project is not a race. The project requires that both sets of activities (or paths) be completed; consequently, the project takes 15 days to complete.

FIGURE 3.16

A Gantt chart compared with a PERT diagram (network diagram) for scheduling activities.

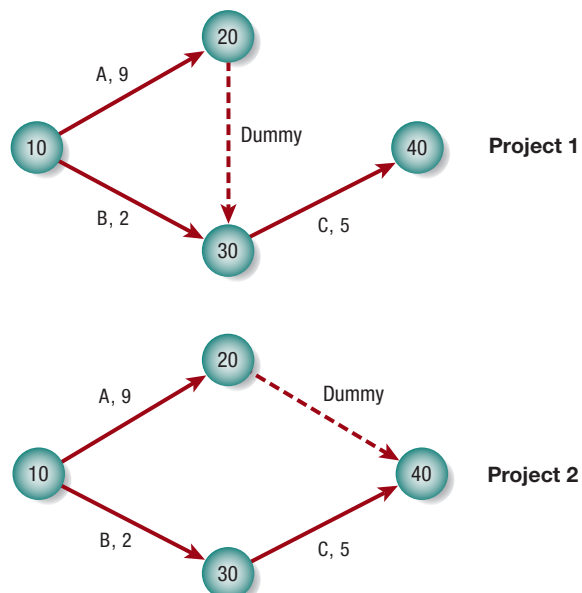


The longest path is referred to as the *critical path*. Although the critical path is determined by calculating the longest path, it is defined as the path that will cause the whole project to fall behind if even one day's delay is encountered on that path. Note that if you are delayed one day on path 10–20–40–50, the entire project will take longer, but if you are delayed one day on path 10–30–40–50, the entire project will not suffer. The leeway to fall behind somewhat on noncritical paths is called *slack time*.

Occasionally, PERT diagrams need pseudo-activities, referred to as dummy activities, to preserve the logic of or to clarify the diagram. Figure 3.17 shows two PERT diagrams with dummies. Project 1 and Project 2 are quite different, and the way the dummy is drawn makes the difference clear. In Project 1, activity C can be started only if both A and B are finished because all arrows coming into a node must be completed before leaving the node. In Project 2, however, activity C requires only activity B's completion and can therefore be under way while activity A is still taking place.

FIGURE 3.17

Precedence of activities is important in determining the length of a project when using a PERT diagram.



Activity	Predecessor	Duration
A Conduct interviews	None	3
B Administer questionnaires	A	4
C Read company reports	None	4
D Analyze data flow	B, C	8
E Introduce prototype	B, C	5
F Observe reactions to prototype	E	3
G Perform cost-benefit analysis	D	3
H Prepare proposal	F, G	2
I Present proposal	H	2

FIGURE 3.18

Listing activities for use in drawing a PERT diagram.

Project 1 takes 14 days to complete, whereas Project 2 takes only 9 days. The dummy in Project 1 is necessary, of course, because it indicates a crucial precedence relationship. The dummy in Project 2, on the other hand, is not required, and activity A could have been drawn from 10 to 40, eliminating event 20 completely.

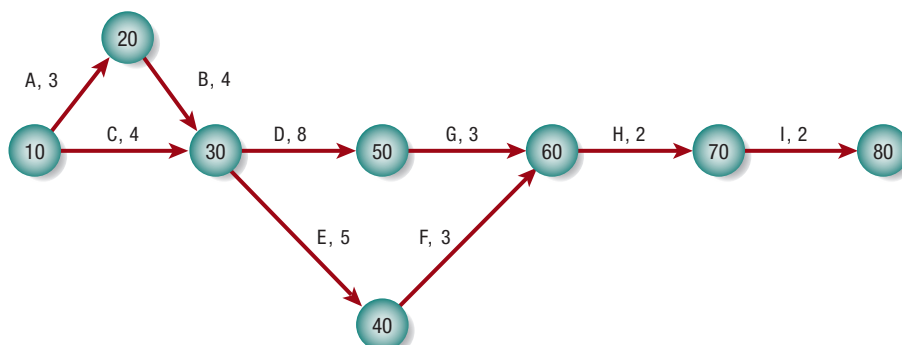
Therefore, there are many reasons for using a PERT diagram over a Gantt chart. The PERT diagram allows:

1. Easy identification of the order of precedence
2. Easy identification of the critical path and thus critical activities
3. Easy determination of slack time

A PERT EXAMPLE Suppose a systems analyst is trying to set up a realistic schedule for the data-gathering and proposal phases of the systems analysis and design life cycle. The systems analyst looks over the situation and lists activities that need to be accomplished along the way. This list, which appears in Figure 3.18, also shows that some activities must precede other activities. The time estimates were determined as discussed in an earlier section of this chapter.

DRAWING THE PERT DIAGRAM In constructing the PERT diagram for this example, the analyst looks first at the activities that require no predecessor activities—in this case A (conduct interviews) and C (read company reports). In the example in Figure 3.19, the analyst chose to number the nodes 10, 20, 30, and so on and drew two arrows out of the beginning node 10. These arrows represent activities A and C and are labeled as such. Nodes numbered 20 and 30 are drawn at the end of these respective arrows. The next step is to look for any activity that requires only A as a predecessor; task B (administer questionnaires) is the only one, so it can be represented by an arrow drawn from node 20 to node 30.

Because activities D (analyze data flow) and E (introduce prototype) require both activities B and C to be finished before they are started, arrows labeled D and E are drawn from node 30, the event that recognizes the completion of both B and C. This process is continued until the entire PERT diagram is completed. Notice that the entire project ends at an event called node 80.

**FIGURE 3.19**

A completed PERT diagram for the analysis phase of a systems project.

IDENTIFYING THE CRITICAL PATH Once the PERT diagram is drawn, it is possible to identify the critical path by calculating the sum of the activity times on each path and choosing the longest path. In this example, there are four paths: 10–20–30–50–60–70–80, 10–20–30–40–60–70–80, 10–30–50–60–70–80, and 10–30–40–60–70–80. The longest path is 10–20–30–50–60–70–80, which takes 22 days. It is essential that the systems analyst carefully monitor the activities on the critical path to keep the entire project on time or even shorten the project length, if warranted.

Controlling a Project

No matter how well a systems analyst plans a project, things can go wrong. In this section, we discuss how to estimate cost and prepare a budget, how to predict and prepare for risk, how to make up time using expediting, and how to control costs when the budget needs revising, the team falls behind, or experiences cost overruns.

Estimating Costs and Preparing the Budget

Keeping a project on time is important, but it is also essential that the cost of a project be managed properly. Once a work breakdown structure is created and a schedule is planned, the analyst needs to:

1. Estimate costs for each activity in the work breakdown structure
2. Prepare a budget for the project and have it approved by the organization or client
3. Manage and control the costs throughout the project

Earlier we discussed cost estimates for equipment and off-the-shelf software, resources we need to complete the project. Now we are concerned with other types of resources needed to complete each task in the work breakdown structure. The main resources for this part of the project are the time of the team members and the type of special equipment and tools needed to finish each of the activities.

Many approaches to cost estimation are available to a systems analyst. They are similar to the estimates used for time estimates and include the following:

1. Basing estimates on similar projects, also called the top-down approach
2. Building bottom-up estimates
3. Using parametric modeling

THE TOP-DOWN APPROACH TO COST ESTIMATION Previous experience with estimating costs means a great deal, especially if the project you are attempting to estimate resembles a project you have previously worked on. If you've developed a website with similar features, then you can reliably estimate the costs of developing a new website.

Customization is possible. The new website may feature a different number of products for sale, but it is possible to adjust the costs according.

THE BOTTOM-UP APPROACH TO COST ESTIMATION Often, an analyst is faced with a project that requires something unique, such as development in a different programming language. In this case, the analyst needs to use a bottom-up approach.

A systems analyst can take the work breakdown structure and ask each responsible project team member to estimate the cost involved for completing the activity that is his or her responsibility. This method, however, yields estimates that may be good or bad, depending on the abilities of each team member. The analyst, acting as a project manager, must review each of these estimates and arrive at a cost estimation that satisfies the team and the client. The obvious problem with the bottom-up method is the time it takes to make each estimate along the way.

PARAMETRIC MODELING This method involves making estimates for each of the many factors, or parameters, that make up a project. For example, you can estimate that it will cost \$75 per line of code and \$80 per hour for the computer programmers needed; you can then estimate the lines of



MAC APPEAL

Color-coding helps a project manager sort out similar phases, tasks, and resources. OmniPlan, available for Macs, takes advantage of color-coding to set up a project, identify tasks, identify the critical path, and flag impossible situations.

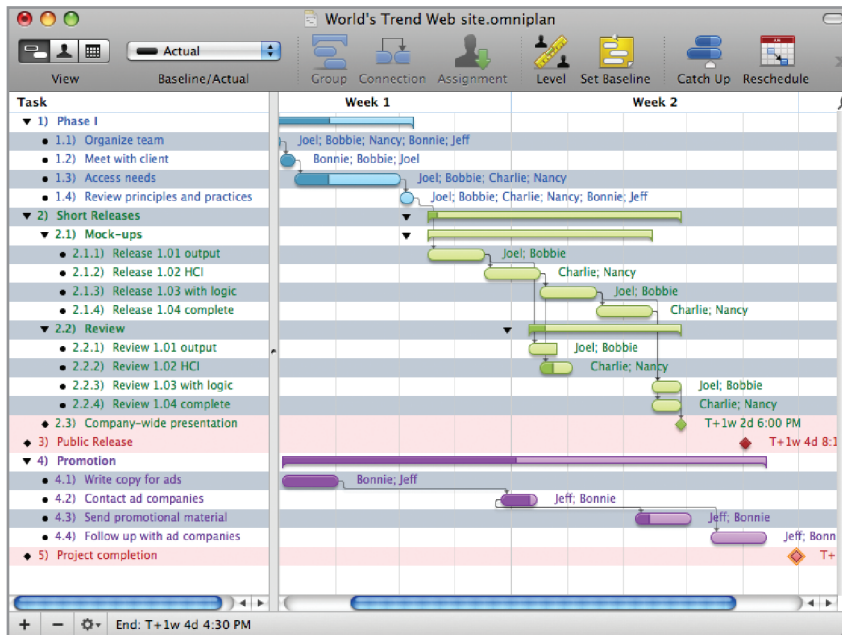


FIGURE 3.MAC

OmniPlan project management software from The Omni Group. (Used by permission of the Omni Group.)

code and hours it will take to complete the project. Special parametric modeling software such as COCOMO II, mentioned previously, is often useful when modeling the project.

WHY COST ESTIMATES FAIL In practice, most systems analysts use a combination of all three of the cost-estimating methods just described. There are two main reasons cost estimates fail. First, an analyst is likely to be overly optimistic. An analyst believes in his or her team members, and an optimistic estimate will show the team completing the project quickly and without errors all the time. An analyst is likely to be optimistic and underestimate the lines of code and effort in general.

Second, an analyst may want to move past the cost estimation process and get on with preparing and presenting the budget and beginning the actual work on the project. The analyst may therefore spend less time than is needed to properly prepare the estimates. The analyst should be as accurate as possible, even though the estimates will be revised as the project continues.

PREPARING THE BUDGET In the end, a systems analyst needs to prepare a budget even if the estimates are not perfect. The budget is a critical deliverable, and every client wants to see a detailed budget early in the process.

When working for a client, a systems analyst needs to conform to the standard process the client uses to develop a budget. Most of the time, the client will have standard forms it uses for this purpose. A sample budget is shown in Figure 3.20.

FIGURE 3.20

Part of a sample budget for a software development project, showing hardware, software, and personnel costs.

Items	Hours or Units	Cost per Hour or Unit	Subtotals
1. Project Team			
Project manager (systems analyst)	600	\$120	\$72,000
Project team members	2,400	80	192,000
Outside contractors (testers)	200	20	4,000
2. Hardware resources			
Workstations	5	4,000	20,000
iPads	5	900	4,500
3. Software			
Off-the-shelf software	5	400	2,000
Software developed in-house		75,000	75,000
4. Training			
Seminars for team members	5	3,500	17,500
Seminars for trainees	5	1,200	6,000
Trainee hourly costs	1,300	20	26,000
Total Project Cost Estimate			\$419,000

The budget contains estimates for the hours worked and the rates of pay for each of the internal or outsourced workers. It also contains hardware and software costs and spells out how much of each type of equipment is needed. It also explains the costs involved with training.

Managing Risk

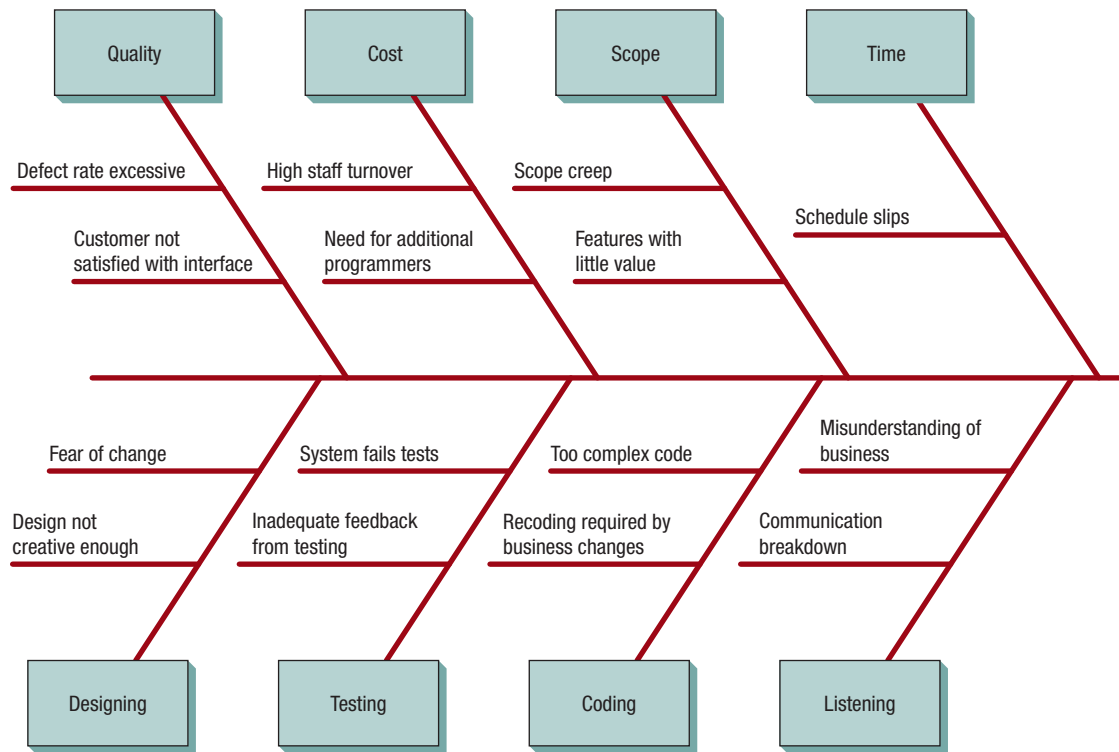
The early discussions you have with management and others requesting a project, along with the feasibility studies you do, are usually the best defenses possible against taking on projects that have a high probability of failure. Your training and experience will improve your ability to judge the worthiness of projects and the motivations that prompt others to request projects. If you are part of an in-house systems analysis team, you must keep current with the political climate of the organization as well as with financial and competitive situations.

It is important, however, to note that systems projects can and do have serious problems. Those that are developed using agile methods are not immune to such troubles. To illustrate what can go wrong in a project, a systems analyst may want to draw a fishbone diagram (also called a *cause-and-effect diagram*, or an *Ishikawa diagram*). When you examine Figure 3.21, you will see why it is called a *fishbone diagram*; it resembles the skeleton of a fish.

When using a fishbone diagram, you systematically list all the problems that can possibly occur. In the case of the agile approach, it is useful to organize a fishbone diagram by listing all the resource control variables on the top and all the activities on the bottom. Some problems, such as schedule slips, might be obvious, but others, such as scope creep (the desire to add features after the analyst hears new stories) or developing features with little value, are not as obvious.

You can learn from the wisdom gained by people involved in earlier project failures. When asked to reflect on why projects had failed, professional programmers cited management setting impossible or unrealistic dates for completion, belief in the myth that simply adding more people to a project would expedite it (even though the original target date on the project was unrealistic), and management behaving unreasonably by forbidding the team to seek professional expertise from outside the group to help solve specific problems.

Remember that you are not alone in the decision to begin a project. Although apprised of your team's recommendations, management will have the final say about whether a proposed project is worthy of further study (that is, further investment of resources). The decision process of your team must be open and stand up to scrutiny from those outside it. The team members should consider that their reputation and standing in the organization are inseparable from the projects they accept.

**FIGURE 3.21**

A fishbone diagram may be used to identify all the things that can go wrong in developing a system.

Managing Time Using Expediting

Speeding up a process is called *expediting*, and sometimes you may benefit from completing a project much faster, such as receiving a bonus if you finish sooner. A second example is realizing that valuable project team resources and team members can be used for some other project if you finish the current project before the due date.

Figure 3.22 lists each activity in the project and the original estimated time required to complete each of these activities. The third column, labeled “crash time,” refers to the absolute minimum time in which an activity can be completed if additional money is funneled to that activity. The final column contains the cost of reducing the activity duration by one week.

Expediting can help reduce the time it takes to complete an entire project, but the expedited activities have to be on the critical path for expediting to make a time difference. Which activity is expedited on the critical path depends on the cost, assuming that the activity is not already at its crash time.

Activity	Estimated Duration	Crash Time	Cost/Week
A	3	1	\$ 800
B	4	2	500
C	4	2	400
D	8	6	1,000
E	5	5	1,000
F	3	3	800
G	3	3	800
H	2	2	400
I	2	1	600

FIGURE 3.22

Estimated duration, crash time, and the cost of expediting project activities.

FIGURE 3.23

Using expediting to minimize project time.

Eligible Activities	Activity Chosen	Time for Each Path				Cost	Cumulative Cost
		22	19	19	16		
A, B, D, or I	B	21	18	19	16	\$ 500	\$ 500
A, B, D, or I	B	20	17	19	15	500	1,000
A, D, or I	I	19	17	18	15	600	1,600
A or D	A	18	16	18	15	800	2,400
A and C, or D	D	17	16	17	15	1,000	3,400
A and C, or D	D	16	16	16	15	1,000	4,400
A and C	A and C	15	15	15	14	1,200	5,600

The maximum number of weeks each activity can be reduced is the difference between the expected time and its crash time. For example, activity B, administering questionnaires, could be reduced from four weeks to two weeks at a cost of \$500 per week, but it cannot be reduced to less than two weeks; activity H, preparing the proposal, cannot be reduced because it is already at its crash time.

The expediting analysis for this example is provided in Figure 3.23. The expediting process takes place one step at a time, until it is impossible to expedite any further. The columns in the table include eligible activities (tasks that are on the critical path and can be reduced by expediting), the activity chosen (because it is the cheapest alternative), the time it currently takes to complete each of the paths, the cost of expediting the chosen activity, and finally the cumulative cost.

In the first step, the critical path is 10–20–30–50–60–70–80, so the eligible activities are A, B, D, and I. Activities G and H are also on the critical path, but they are already at their crash times and are consequently ineligible for expediting. The cheapest alternative is to expedite activity B by one day, which reduces the first path from 22 to 21 weeks and the second from 19 to 18 weeks; the third and fourth paths are not affected by the reduction because activity B is not on either of those paths.

The critical path, and therefore the entire project, is reduced from 22 to 21 weeks (circled on the table). We can repeat this reduction and reduce the project time by another week.

When activity B reaches its crash time, another activity must be chosen. Row 3 in the table shows that activities A, D, and I are eligible, and activity I is the cheapest alternative. Reducing activity I reduces not only the critical path but all paths because it is common to all of them.

In the fourth step, activity A is chosen, reducing paths 1 and 2, but as a result, there are now two critical paths. This implies that any reduction of the project time will take place only if both of the critical paths are reduced at the same time.

We can shorten both paths in the fifth and sixth steps by choosing either a combination of activities A and C (one activity from each of the critical paths) or activity D (an activity common to both critical paths). Reducing activity D by 2 days shortens the paths to 16, 16, 16, and 15 days, respectively, and now there are three critical paths.

Finally, when activity D reaches its crash time, the only available choice is a combination of activities A and C. The minimum project time is therefore 15 weeks, obtainable by reducing activity A by 2 days, activity B by 2 days, activity C by 1 day, activity D by 2 days, and activity I by one day, at a total cost of \$5,600.

This example describes all-out expediting to obtain the minimum project time at any cost. But a systems analyst may be faced with a budget. In our example, a budget of \$4,000 would result in expediting up to and including step 5. The project would be shortened from 22 to 17 weeks, at a cost of \$3,400.

Another possible criterion would be the net amount that could be saved if the project were shortened. Suppose that in the above example the analyst would save \$750 per week, mostly consisting of opportunities available for the project team to begin new projects sooner. In this case, expediting would not take place until Step 3 because the incremental cost of Step 4 (\$800 for expediting activity A) would exceed the \$750 saved.

Expediting can make or break a successful project. The systems analyst has to remain on top of the situation by managing the project throughout the entire development process. The analyst needs to make sure project costs are managed properly.

Controlling Costs Using Earned Value Management

Once a systems analyst has an approved budget, it is imperative that the analyst keeps it updated so the project doesn't exceed the budget as it progresses. The baseline needs to be continually revised, and all the stakeholders need to be informed.

Changes often occur in the middle of a project. The client may request new features, or new technologies may be introduced that will change the way the system is developed. No matter what these changes are, the budget needs to be revised.

One tool available to systems analysts is earned value management (EVM). It is a technique used to help determine progress (or setbacks) on a project and involves project cost, the project schedule, and the performance of the project team. Here are the four key measures in EVM:

- *Budget at completion (BAC)* is the total budget for the project, from the beginning until completion. If you are calculating the performance measures for a task, then it is the total budget for the task.
- *Planned value (PV)* is the value of the work that is to be completed on the project (or, alternatively, the work completed on any task). Because the value of the work completed is how much effort and money will be put into it, you can think of PV as the budgeted cost of work scheduled.
- *Actual Cost (AC)* is the total cost (direct and indirect) incurred in completing the work on the project (or, alternatively, a task) up to that particular point in time. Another way of referring to this is actual cost of work performed to date.
- *Earned value (EV)* is an estimate of the value of the work performed thus far. EV refers only to the work that has been completed to date. Earned value is calculated as follows:

$$EV = PV * p$$

where p is the percentage of the work completed thus far.

Figure 3.24 shows the cost of developing a website over a five-month period. In this example, the development of our website is well under way. The budget at completion is the planned budget of \$18,000, the total estimated cost of developing the website at the end of five months. It is calculated by summing the cost figures in the estimated cost column.

At the end of the first, second, and third months, the actual cost of the project equals the cumulative estimated cost, but in the fourth month, the actual cost is \$17,000 versus the \$15,000 from the cumulative estimate column. It is obvious that we are going over our budget.

It appears to be worse. We've only completed 50 percent of Stage 4 at the end of the fourth month. That means we're falling behind in our schedule as the costs are rising.

Let's see what the damage is. We already know that the budget at completion for the project is \$18,000 and that the actual cost at the end of the fourth month is \$17,000. Let's determine the planned value and the earned value at this time.

The planned value is simply the cumulative estimate at the end of month 4, or \$15,000. To calculate the planned value we need to multiply the earned value by the percentage of the project

At the End of	Stage	Estimated Cost	Cumulative Estimate	Estimated Duration	Stage Completed	Actual Cost of Stage to Date	Actual Cost of Project to Date
Month 1	Stage 1	\$6,000	\$6,000	1 month	100%	\$6,000	\$6,000
Month 2	Stage 2	3,000	9,000	1 month	100%	3,000	9,000
Month 3	Stage 3	3,000	12,000	1 month	100%	3,000	12,000
Month 4	Stage 4	3,000	15,000	1 month	50%	5,000	17,000
Month 5	Stage 5	3,000	18,000	1 month	0%	Not yet begun	Not yet begun

FIGURE 3.24

The anticipated cost of a website development project over a five-month period.

that is complete. We see in the table that the first three stages are complete, but the fourth stage is only half finished. To date, this means that:

$$p = (100 + 100 + 100 + 50) / (100 + 100 + 100 + 100) \\ = .875$$

So the earned value is:

$$EV = PV * p \\ = \$15,000 * .875 \\ = \$13,125$$

The four variables—budget at completion, planned value, actual cost, and earned value—are the core of earned value management. They can be used to calculate a number of performance measures, including cost variance, schedule variance, the cost performance index, and the schedule performance index:

- *Cost variance (CV)* tells us whether the project or task is costing more than we have planned. If cost variance is negative, our costs are greater than planned. If it is positive, it is costing less than planned. Cost variance is simply:

$$CV = EV - AC$$

- *Schedule variance (SV)* tells us whether the project is taking more time than planned. If so, it will have a negative impact on our costs. Once again, if schedule variance is negative, then it is taking longer than planned to perform the work, and the amount is the cost overage. If schedule variance is positive, then it is taking less time than planned, and the amount indicates how much we are under budget. Schedule variance is obtained by subtracting the planned value from the earned value, as follows:

$$SV = EV - PV$$

- *Cost performance index (CPI)* is a ratio that signifies whether a project is over budget. If the index is less than 1.0, the project or task is over budget. If the ratio is greater than 1.0, the project or task is under budget. Cost performance index can be calculated as follows:

$$CPI = EV / AC$$

- *Schedule performance index (SPI)* is a ratio that tells us whether a project is falling behind schedule. If the schedule performance index is less than 1.0, it implies that the project or task is behind schedule. If the schedule performance index ratio is greater than 1.0, the project is ahead of schedule. Schedule performance index can be calculated as follows:

$$SPI = EV / PV$$

In our continuing example of website development, we can now calculate these four performance measures:

Cost variance at the end of the fourth month is:

$$CV = EV - AC \\ = \$13,125 - \$17,000 \\ = (-\$3,875)$$

meaning that we are \$3,875 over budget.

Schedule variance is:

$$SV = EV - PV \\ = \$13,125 - \$15,000 \\ = (-\$1,875)$$

implying that we are behind schedule and are over budget.

The cost performance index is:

$$\begin{aligned}\text{CPI} &= \text{EV} / \text{AC} \\ &= \$13,125 / \$17,000 \\ &= .772\end{aligned}$$

or 77.2 percent. This means the team is performing at a rate far below the expected rate of performance. This figure will be used later.

The schedule performance index (SPI) is:

$$\begin{aligned}\text{SPI} &= \text{EV} / \text{PV} \\ &= \$13,125 / \$15,000 \\ &= .875\end{aligned}$$

or 87.5 percent. Once again, a ratio lower than 1.0 means that the project is behind schedule.

Finally, a systems analyst may want to calculate two key variables: estimate to complete (ETC) and estimate at completion (EAC). They tell the analyst how much it will cost to finish the project or task from this point onward.

Estimate to complete considers how the team has done in the past and estimates how much additional money will be required to complete the project at its current rate of performance. It is calculated by subtracting the earned value from the budget at completion. That result is divided by the cost performance index calculated earlier, as follows:

$$\text{ETC} = (\text{BAC} - \text{EV}) / \text{CPI}$$

Estimate at completion represents the revised budget. It is the amount the entire project or task will cost when it is finished. It is calculated by adding the actual cost to the estimated amount of money it will take to complete the project. The equation is:

$$\text{EAC} = \text{AC} + \text{ETC}$$

In our ongoing example of the website development project, the estimate to complete the project is:

$$\text{ETC} = (\$18,000 - \$13,125) / .772 = \$6,315$$

This means that we will need to spend \$6,315 more than we have up to this point to finish the website. The estimate at completion will be:

$$\begin{aligned}\text{EAC} &= \$17,000 + \$6,315 \\ &= \$23,315\end{aligned}$$

which is significantly more than the \$18,000 we originally budgeted for. Figure 3.25 shows how some of these performance measures relate to the original schedule and budget and the revised schedule and budget.

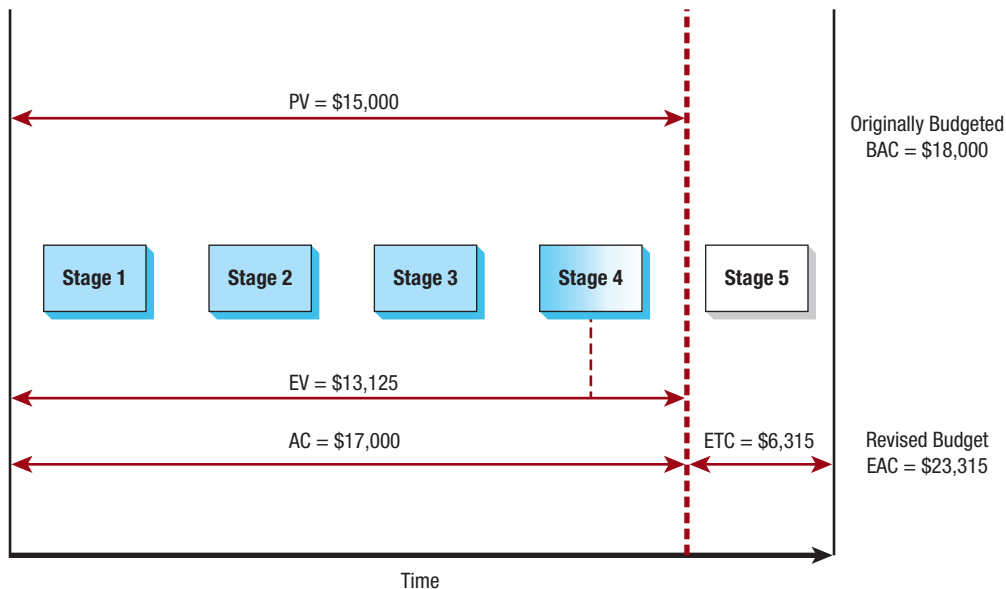


FIGURE 3.25

How the earned value management values relate to the planned and actual development of the website.

It is important for a systems analyst to keep an eye on cost and to address the budgetary implications when unexpected delays or cost overruns occur. An analyst must constantly balance cost, time, and scope (that is, which systems features will be included) when working on any project.

Managing the Project Team

Along with managing time and resources, systems analysts must also manage people. Management is accomplished primarily by communicating accurately to team members who have been selected for their competence and compatibility. Goals for project productivity must be set, and members of systems analysis teams must be motivated to achieve them.

Assembling a Team

Assembling a team is desirable. If a project manager has an opportunity to create a dream team of skilled people to develop a system, who should he or she choose? In general, project managers need to look for others who share their values of teamwork and are guided by the desire to deliver a high-quality system on time and on budget. Other desirable team member characteristics include a good work ethic, honesty, competence, and readiness to take on leadership based on expertise, motivation, enthusiasm for the project, and trust of teammates.

The project manager needs to know about business principles, but it doesn't hurt to have at least one other person on the team who understands how a business operates. Perhaps this person should be a specialist in the area of the system being developed. When developing an ecommerce site, for example, a team can enlist the help of someone in marketing, and those developing an inventory system can ask a person versed in production and operations to provide expertise.

A team ideally should include at least two systems analysts. They can help each other, check each other's work, and shift their workloads. There is certainly a need to have people with programming skills on board. Coding is important, but people who know how to conduct walkthroughs, reviews, testing, and documenting systems are important as well. Some people are good at seeing the big picture, and others perform well when tasks are broken down into smaller ones for them. Every team should have both types of individuals.

Beyond the basics just described, a project manager should look for people with both experience and enthusiasm. Experience is especially important when trying to estimate the time required for completing a project. Experience in computer programming can mean code is developed five times faster than if it is developed by an inexperienced team. A usability expert is also a useful addition to the team.

The team must be motivated. One way to keep a team positively oriented throughout the entire process is to select good people at the outset. Look for enthusiasm, imagination, and an ability to communicate with different kinds of people. These basic attributes hold the potential for success. In addition, superior writers and articulate speakers can present proposals and work directly with customers, so hiring such individuals is a good idea.

Trust is an important part of a team. All members of the project need to act responsibly and agree to do their best and complete their part of the project. People may have different work styles, but they all need to agree to work together toward a common goal.

Communication Strategies for Managing Teams

Teams have their own personalities as a result of combining each individual team member with every other in a way that creates a totally new network of interactions. A way to organize your thinking about teams is to visualize them as always seeking a balance between accomplishing the work at hand and maintaining relationships among team members.

In fact, teams often have two leaders, not just one. Usually one person will emerge who leads members to accomplish tasks, and another person will emerge who is concerned with



HYPERCASE EXPERIENCE 3.1

“I hope everyone you’ve encountered at MRE has treated you well. Here’s a short review of some of the ways you can access our organization through HyperCase. The reception area at MRE contains the key links to the rest of our organization. Perhaps you’ve already discovered these on your own, but I wanted to remind you of them now because I don’t want to get so engrossed in the rest of our organization problems that I forget to mention them.

“The empty doorway you see is a link to the next room, which we call the East Atrium. You have probably noticed that all open doorways are links to adjacent rooms. Notice the building map displayed in the reception area. You are free to go to public areas such as the canteen, but as you know, you must have an employee escort you into a private office. You cannot go there on your own.

“By now you have probably noticed the two documents and the computer on the small table in the reception area. The little red one is the MRE internal phone directory. Just click on an employee name, and if that person is in, he or she will grant you an interview and a tour of the office. I leave you to your own devices in figuring out what the other document is.

“The computer on the table is on and displays the web page for MRE. Take a look at the corporate website and visit all the links.

It tells the story of our company and the people who work here. We’re quite proud of it and have gotten positive feedback about it from visitors.

“If you have had a chance to interview a few people and see how our company works, I’m sure you are becoming aware of some of the politics involved. We are also worried, though, about more technical issues, such as what constitutes feasibility for a training project and what does not.”

HYPERCASE Questions

1. What criteria do the Training Unit use to judge the feasibility of a new project? List them.
2. List any changes or modifications to these criteria that you would recommend.
3. Mr. Evans has asked you to help prepare a proposal for a new project tracking system for the Training Unit. Briefly discuss the technical, economic, and operational feasibility of each alternative for a proposed project tracking system for the Training Unit.
4. Which option would you recommend? Use evidence from HyperCase to support your decision.

the social relationships among group members. Both are necessary for the team. These individuals have been labeled by other researchers as task leader and socioemotional leader, respectively. Every team is subject to tensions that are an outgrowth of seeking a balance between accomplishing tasks and maintaining relationships among team members.

For a team to continue its effectiveness, tensions must be continually resolved. Minimizing or ignoring tensions will lead to ineffectiveness and eventual disintegration of the team. Much of the tension release necessary can be gained through skillful use of feedback by all team members. All members, however, need to agree that the way they interact (that is, the process) is important enough to merit some time. Productivity goals for processes are discussed in a later section.

Securing agreement on appropriate member interaction involves creating explicit and implicit team norms (collective expectations, values, and ways of behaving) that guide members in their relationships. A team’s norms belong to it and will not necessarily transfer from one team to another. These norms change over time and are better thought of as a team process of interaction than a product.

Norms can be functional or dysfunctional. Just because a particular behavior is a norm for a team does not mean it is helping the team to achieve its goals. For example, an expectation that junior team members should do all project scheduling may be a team norm. By adhering to this norm, the team puts extreme pressure on new members and does not take full advantage of the experience of the team. It is a norm that, if continued, could make team members waste precious resources.

Team members need to make norms explicit and periodically assess whether norms are functional or dysfunctional in helping the team achieve its goals. The overriding expectation for your team must be that change is the norm. Ask yourself whether team norms are helping or hindering the team’s progress.

Setting Project Productivity Goals

As you work with your team members on a variety of projects, you or your team leader will acquire acumen for projecting what the team can achieve in a specific amount of time. Using the hints on methods for estimating time required discussed earlier in this chapter and coupling them with experience will enable the team to set worthwhile productivity goals.

Systems analysts are accustomed to thinking about productivity goals for employees who show tangible outputs, such as the number of pairs of blue jeans sewn per hour, the number of entries keyed in per minute, or the number of items scanned per second. As manufacturing productivity rises, however, it is becoming clear that managerial productivity must keep pace. It is with this aim in mind that productivity goals for the systems analysis team are set.

A team needs to formulate and agree to goals, and these goals should be based on team members' expertise and former performance, as well as on the nature of the specific project. Goals will vary somewhat for each project undertaken because sometimes an entire system will be installed, whereas other projects might involve limited modifications to a portion of an existing system.

Motivating Project Team Members

Although motivation is an extremely complex topic, it is a good one to consider, even if briefly, at this point. To oversimplify, recall that people join organizations to provide for some of their basic needs, such as food, clothing, and shelter. All humans, however, also have higher-level needs, including affiliation, control, independence, and creativity. People are motivated to fulfill unmet needs on several levels.

Team members can be motivated, at least partially, through participation in goal setting, as described in the previous section. The very act of setting a challenging but achievable goal and then periodically measuring performance against the goal seems to work in motivating people. Goals act almost as magnets in attracting people to achievement.

Part of the reason goal setting motivates people is that team members know prior to any performance review exactly what is expected of them. The success of goal setting for motivating can also be ascribed to affording each team member some autonomy in achieving the goals. Although a goal is predetermined, the means to achieve it may not be. In this instance, team members are free to use their own expertise and experience to meet their goals.

Setting goals can motivate team members by clarifying for them and others what must be done to get results. Goals define the level of achievement that is expected of team members. This use of goals simplifies the working atmosphere, and it electrifies it with the possibility that what is expected can indeed be done.

Managing Ecommerce Projects

Many of the approaches and techniques discussed so far are transferable to ecommerce project management. You should be cautioned, however, that although there are many similarities, there are also many differences. One difference is that the data used by ecommerce systems are scattered all over an organization. Therefore, you are not just managing data in a self-contained department or even in one solitary unit. Hence, organization politics may come into play because units often feel protective of the data they generate and do not understand the need to share them across the organization.

Another stark difference is that ecommerce project teams typically need more staff from across the organization and with a variety of skills, including developers, consultants, database experts, and system integrators. Stable project groups that exist within a cohesive information systems (IS) group or systems development team are the exception rather than the rule. In addition, because so much help may be required initially, ecommerce project managers need to build partnerships externally and internally well ahead of the implementation, perhaps sharing talent across projects to defray costs of ecommerce implementations and to muster the required number of people with the necessary expertise. The potential for organization politics to drive a wedge between team members is very real.

One way to prevent politics from sabotaging a project is for the ecommerce project manager to emphasize the integration of the ecommerce system with the organization's internal systems and in so doing emphasize the organizational aspect embedded in the ecommerce project. As one ecommerce project manager told us, "Designing the front end [what the consumer sees] is the easy part of all this. The real challenge comes from integrating ecommerce strategically into all of the organization's systems."

A fourth difference between traditional project management and ecommerce project management centers on security concerns. The ecommerce system will be linking with the outside world via the Internet, so security is of the utmost importance. Developing and implementing a security plan before the new system is in place is a project in and of itself and must be managed as such.

Creating a Project Charter

Part of the planning process is to agree on what will be done and at what time. Analysts who are external consultants, as well as those who are organization members, need to specify what they will eventually deliver and when they will deliver it. The entire team as well as the client need to be on board. This chapter has elaborated on ways to estimate the delivery date for the completed system and also how to identify organization goals and assess the feasibility of the proposed system.

The project charter is a written narrative that clarifies the following questions:

1. What does the user expect of the project (that is, what are the objectives)? What will the system do to meet the needs (achieve the objectives)?
2. What is the scope (or what are the boundaries) of the project? (What does the user consider to be beyond the project's reach?)
3. What analysis methods will the analyst use to interact with users in gathering data and developing and testing the system?
4. Who are the key participants? How much time are users willing and able to commit to participating?
5. What are the project deliverables? (What new or updated software, hardware, procedures, and documentation do the users expect to have available for interaction when the project is done?)
6. Who will evaluate the system, and how will they evaluate it? What are the steps in the assessment process? How will the results be communicated, and to whom?
7. What is the estimated project timeline? How often will analysts report project milestones?
8. Who will train the users?
9. Who will maintain the system?

The project charter describes in a written document the expected results of the systems project (deliverables) and the time frame for delivery. It essentially becomes a contract between the chief analyst (or project manager) and the analysis team, with the organization users requesting the new system.

The Systems Proposal

The project charter serves the purpose of identifying objectives, determining scope, and assigning responsibilities, but an analyst still needs to prepare a systems proposal that includes much of the detail about system needs, options, and recommendations. This section covers both the content and style that make up a systems proposal.

What to Include in a Systems Proposal

A written systems proposal has 10 main sections. Each section has a particular function, and eventually the proposal should be arranged in the following order:

1. Cover letter
2. Title page of project



CONSULTING OPPORTUNITY 3.5

Goal Tending

“Here’s what I think we can accomplish in the next five weeks,” says Hy, the leader of your systems analysis team, as he confidently pulls out a schedule listing each team member’s name alongside a list of short-term goals. Just a week ago, your systems analysis team went through an intense meeting on expediting the project schedule for the Kitchener Redwings, an Ontario hockey organization whose management is pressuring you to produce a prototype.

The three other members of the team look at the chart in surprise. Finally, one of the members, Rip, speaks: “I’m in shock. We each have so much to do as it is—and now this.”

Hy replies defensively, “We’ve got to aim high, Rip. They’re in the off-season. It’s the only time to get them. If we set our goals too low, we won’t finish the system prototype, let alone the system itself, before another hockey season passes. The idea is to give the Kitchener Redwings the fighting edge through the use of their new system.”

Fiona, another team member, enters the discussion, saying, “Goodness knows their players can’t give them that!” She pauses

for the customary groan from the assembled group, then continues. “But seriously, these goals are killers. You could have at least asked us what we thought, Hy. We may even know better than you what’s possible.”

“This is a pressing problem, not a tea party, Fiona,” Hy replies. “Polite polling of team members was out of the question. Something had to be done quickly. So I went ahead with these. I say we submit our schedule to management based on this. We can push back deadlines later if we have to. But this way, they’ll know we’re committed to accomplishing a lot during the off-season.”

As a fourth team member listening to the foregoing exchange, formulate three suggestions that would help Hy improve his approach to goal formation and presentation. How well motivated do you think the team will be if they share Fiona’s view of Hy’s goals? What are the possible ramifications of supplying management with overly optimistic goals? Write one paragraph devoted to short-term effects and another one discussing the long-term effects of setting unrealistically high goals.

3. Table of contents
4. Executive summary (including recommendations)
5. Outline of systems study with appropriate documentation
6. Detailed results of the systems study
7. Systems alternatives (three or four possible solutions)
8. Systems analysts’ recommendations
9. Proposal summary
10. Appendices (assorted documentation, summary of phases, correspondence, and so on)

A cover letter to managers and the IT task force should accompany the systems proposal. It should list the people who did the study and summarize the objectives of the study. Keep the cover letter concise and friendly.

Include on the title page the name of the project, the names of the systems analysis team members, and the date the proposal is submitted. The proposal title must accurately express the content of the proposal, but it can also exhibit some imagination. The table of contents can be useful to readers of long proposals. If the proposal is less than 10 pages long, omit the table of contents.

The executive summary, in 250 to 375 words, provides the who, what, when, where, why, and how of the proposal, just as would the first paragraph in a news story. It should also include the recommendations of the systems analysts and desired management action because some people will only have time to read the summary. It should be written last, after the rest of the proposal is complete.

The outline of the systems study provides information about all the methods used in the study and who or what was studied. Any questionnaires, interviews, sampling of archival data, observation, or prototyping used in the systems study should be discussed in this section.

This detailed results section describes what the systems analyst has found out about human and systems needs through all the methods described in the preceding section. Conclusions about problems workers experience when interacting with technologies and systems that have come to the fore through the study should be noted here. This section should list the problems or suggest opportunities that call forth the alternatives presented in the next section.

In the systems alternatives section of the proposal, present two or three alternative solutions that directly address the aforementioned problems. The alternatives you present should include one that recommends keeping the system the same. Each alternative should be explored separately. Describe the costs and benefits of each situation. Because there are usually trade-offs involved in any solution, be sure to include the advantages and disadvantages of each.

Each alternative must clearly indicate what users and managers must do to implement it. The wording should be as clear as possible, such as “Buy notebook computers for all middle managers,” “Purchase packaged software to support users in managing inventory,” or “Modify the existing system through funding in-house programming efforts.”

After the systems analysis team has weighed the alternatives, it will have a definite professional opinion about which solution is most workable. The systems analysts’ recommendations section expresses the *recommended* solution. Include the reasons supporting the team’s recommendation so that it is easy to understand why it is being made. The recommendation should flow logically from the preceding analysis of alternative solutions, and it should clearly relate the human–computer interaction findings to the choice offered.

The proposal summary is a brief statement that mirrors the content of the executive summary. It gives the objectives of the study and the recommended solution. You should once more stress the project’s importance and feasibility, along with the value of the recommendations for reaching the users’ goals and improving the business. Conclude the proposal on a positive note.

The appendices are the last part of the systems proposal, and they can include any information that you feel may be of interest to specific individuals but that is not essential for understanding the systems study and what is being proposed.

Once the systems proposal is written, carefully select who should receive the report. Personally hand the report to the people you have selected. Your visibility is important for the acceptance and eventual success of the system.

You can propose a special meeting of key decision makers regarding the systems project for the purpose of delivering the results of the systems proposal. Develop a separate oral presentation (illustrated with PowerPoint slides or some other presentation software such as Keynote) that features highlights of your written report. Keep your presentation brief, 30 to 40 minutes maximum, with the majority of time devoted to questions. Never read your report aloud. It is preferable to prepare and deliver a dynamic slide presentation that takes advantage of the fact that you can interact with key decision makers face to face.

Using Figures for Effective Communication

The emphasis so far in this section has been on considering your audience when composing a systems proposal. Tables and graphs as well as words are important in capturing and communicating the basics of the proposed system. Never underestimate good design.

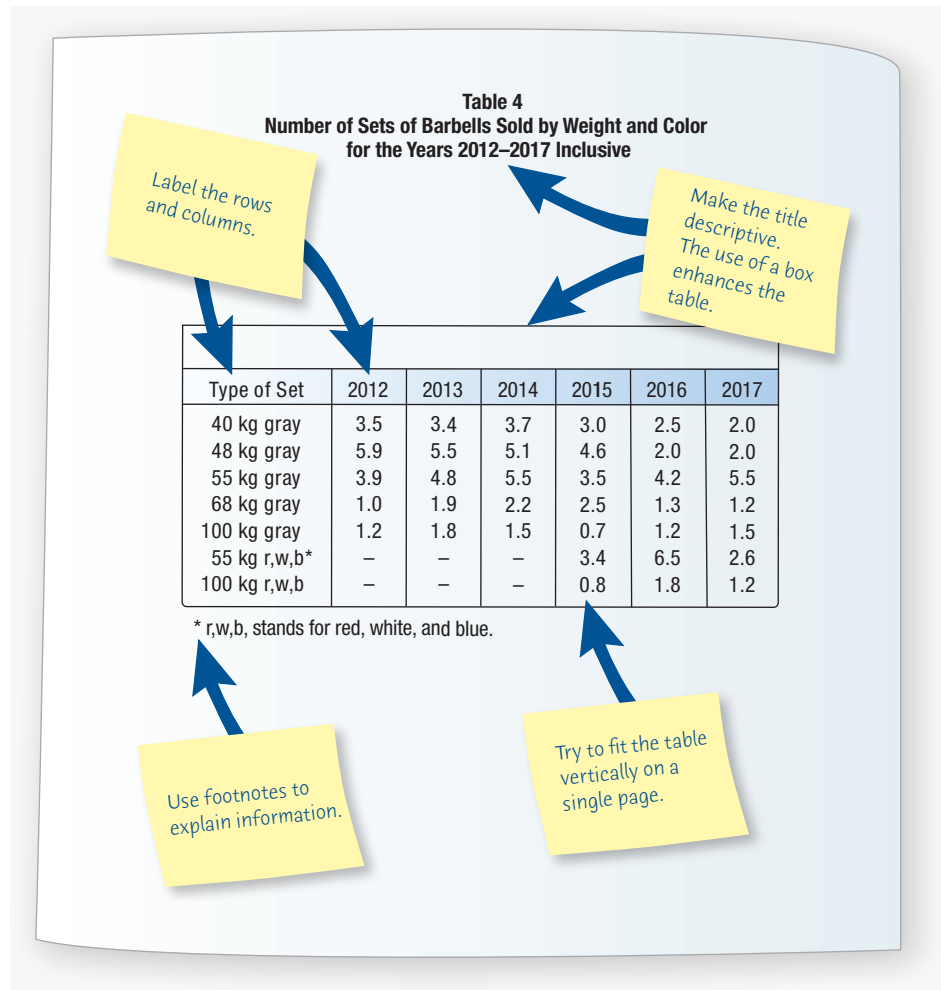
Integrating figures into your proposal helps demonstrate that you are responsive to the different ways people absorb information. Figures in the report supplement written information and must always be interpreted in words; they should never stand alone.

EFFECTIVE USE OF TABLES Although tables are technically not visual aids, they provide a different way of grouping and presenting analyzed data that an analyst wants to communicate to proposal readers.

Tables use labeled columns and rows to present statistical or alphabetical data in an organized way. Each table must be numbered according to the order in which it appears in

FIGURE 3.26

Guidelines for creating effective tables.



the proposal and should be meaningfully titled. Figure 3.26 shows the appropriate layout and labeling for a table.

The following are some guidelines for tables:

1. Integrate tables into the body of the proposal. Don't relegate them to the appendices.
2. Try to fit an entire table vertically on a single page, if possible.
3. Number and title a table at the top of the page. Make the title descriptive and meaningful.
4. Label each row and column. Use more than one line for a title, if necessary.
5. Use a boxed table if room permits. Vertically ruled columns will enhance readability.
6. Use footnotes if necessary to explain detailed information contained in the table.

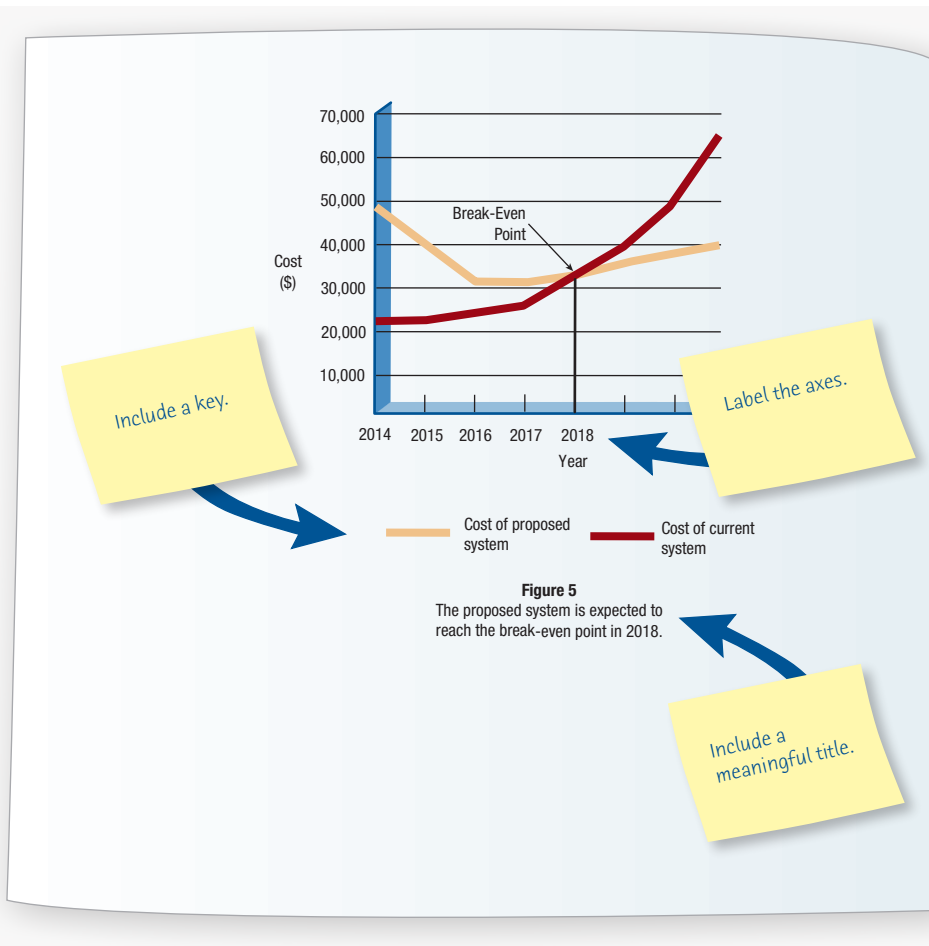
Several methods for comparing costs and benefits were presented in previous sections. Tabled results of such comparisons should appear in the systems proposal. If a break-even analysis is done, a table illustrating results of the analysis should be included. Payback can be shown in tables that serve as additional support for graphs. A short table comparing computer systems or options might also be included in the systems proposal.

EFFECTIVE USE OF GRAPHS There are many different kinds of graphs, including line graphs, column graphs, bar charts, pie charts, and area charts. Line graphs, column graphs, and bar charts compare variables, whereas pie charts and area charts illustrate the percentage composition of an entity.

The following guidelines will help you include effective graphs in a proposal (see Figure 3.27):

FIGURE 3.27

Guidelines for developing effective line graphs.



1. Choose a style of graph that communicates your intended meaning well.
2. Integrate the graph into the body of the proposal.
3. Give the graph a sequential figure number and a meaningful title.
4. Label each axis and any lines, columns, bars, or pieces of the pie on the graph.
5. Include a key to indicate differently colored lines, shaded bars, or crosshatched areas.

Much of the detail that goes into a systems proposal is obtained through interviews, questionnaires, sampling, discovery of other hard data, and observation. These topics are discussed in the next two chapters.



HYPERCASE EXPERIENCE 3.2

“Sometimes the people who have been here for some time are surprised at how much we have actually grown. Yes, I do admit that it isn’t easy to keep track of what each person is up to or even what purchases each department has made in the way of hardware and software. We’re working on it, though. Mr. Evans would like to see more accountability for computer purchases. He wants to make sure we know what we have, where it is, why we have it, who’s using it, and if it’s boosting MRE productivity, or, as he so delicately puts it, ‘to see whether it’s just an expensive toy’ that we can live without.”

HYPERCASE Questions

1. Complete a computer equipment inventory for the Training and Management Systems Department, describing all the systems you find. *Hint:* Create an inventory form to simplify your task.
2. Using the software evaluation guidelines given in the chapter, do a brief evaluation of GEMS, a software package used by the Training and Management Systems Department employees. In a paragraph, briefly critique this custom software by comparing it with COTS software such as Microsoft Project.
3. List the intangible costs and benefits of GEMS, as reported by employees of MRE.
4. Briefly describe the two alternatives Mr. Evans is considering for the proposed project tracking and reporting system.
5. What organizational and political factors should Mr. Evans consider in proposing his new system at MRE? (In a brief paragraph, discuss three central conflicts.)



FIGURE 3.HC1

The reception room resembles a typical corporation. While you are in this HyperCase screen, click on the small red directory if you want to visit an MRE employee.

Summary

The five major project management fundamentals a systems analyst must handle are (1) project initiation—defining a problem, (2) determining project feasibility, (3) activity planning and control, (4) project scheduling, and (5) managing systems analysis team members. When faced with questions of how businesses can

meet their goals and solve systems problems, an analyst creates a problem definition. A problem definition is a formal statement of the problem, including (1) the issues of the present situation, (2) the objectives for each issue, (3) the requirements that must be included in all proposed systems, and (4) the constraints that limit system development.

Selecting a project is a difficult decision because more projects will be requested than can actually be done. Five important criteria for project selection are (1) that the requested project be backed by management, (2) that it be timed appropriately for a commitment of resources, (3) that it move the business toward attainment of its goals, (4) that it be practical, and (5) that it be important enough to be considered over other possible projects.

If a requested project meets these criteria, a feasibility study of its operational, technical, and economic merits can be done. Through the feasibility study, a systems analyst gathers data to enable management to decide whether to proceed with a full systems study. By inventorying equipment already on hand and on order, systems analysts can better determine whether new, modified, or current computer hardware is to be recommended.

There are advantages and disadvantages to purchasing computer hardware. Alternatively the analyst may help the organization weigh the advantages and disadvantages of renting time and space through use of a cloud service provider. Three main categories of cloud computing are Software as a Service (SaaS), Infrastructure as a Service (IaaS), and Platform as a Service (PaaS). As user sophistication grows, some companies are adopting bring your own device (BYOD) approaches to boost morale of employees and lower the initial cost to the organization of purchasing mobile devices. Security risks from untrained users are likely the biggest threat to organizations adopting a “bring your own device” policy.

Cloud services and other hardware vendors will supply support services such as preventive maintenance and user training that are typically negotiated separately. Software can be created as a custom product, purchased as a commercial off-the-shelf (COTS) software package, or outsourced to a SaaS provider.

Preparing a systems proposal means identifying all the costs and benefits of a number of alternatives. A systems analyst has a number of methods available to forecast future costs, benefits, volumes of transactions, and economic variables that affect costs and benefits. Costs and benefits can be tangible (quantifiable) or intangible (nonquantifiable and resistant to direct comparison). A systems analyst can use many methods for analyzing costs and benefits, including break-even analysis and the payback method.

Project planning includes the estimation of time required for each of the analyst’s activities, scheduling the activities, and expediting them if necessary to ensure that a project is completed on time.

To complete projects on time, within budget, and with the features that are promised, a project needs to be broken down into smaller tasks or activities. These tasks together make up a work breakdown structure (WBS), usually through decomposition. There are different types of work breakdown structures; they can be product oriented or process oriented. A process-oriented WBS is typical in systems analysis and design.

One technique available to a systems analyst for scheduling tasks is a Gantt chart, which displays activities as bars on a graph. Another technique, called Program Evaluation and Review Technique (PERT), displays activities as arrows on a network. PERT charts are also called network diagrams. PERT helps an analyst determine the critical path and slack time, which is the information required for effective project control. Both Gantt charts and PERT charts can be developed with Microsoft Project.

Earned value management (EVM) is a technique used to help the analyst determine progress (or setbacks) on a project and involves project cost, the project schedule, and the performance of the project team. The four key measures in EVM are budget at completion (BAC), planned value (PV), actual cost (AC), and earned value (EV). These four measures or variables are the core of EVM. They can be used to calculate a number of performance measures, including cost variance (CV), schedule variance (SV), the cost performance index (CPI), and the schedule performance index (SPI).

Creating a project charter that contains user expectations and analyst deliverables is recommended. Reasons computer programmers cited for projects that had failed included unrealistic management deadlines, adding unneeded personnel to a project that is trying to meet an unrealistic deadline, and not permitting developer teams to seek expert help outside their immediate group. Project failures can usually be avoided by examining the motivations for requested projects and understanding a team’s motives for recommending or avoiding a particular project.

A systems analyst must follow these steps when putting together an effective systems proposal: Effectively organize the proposal content and write and deliver the proposal in an appropriate business style. To be effective, a proposal should be written in a clear and understandable manner, and its content should be divided into 10 functional sections. Visual considerations are important when putting together a proposal. Face-to-face meetings with key decision makers provide an opportunity to orally present the systems proposal in a brief and compelling way.

Keywords and Phrases

actual cost (AC)	intangible costs
benchmarking	moving average
break-even analysis	operational feasibility
bring your own device (BYOD)	payback
bring your own technology (BYOT)	planned value (PV)
budget	Program Evaluation and Review Technique (PERT)
budget at completion (BAC)	problem definition
cloud computing	productivity goals
cost performance index (CPI)	project charter
cost variance (CV)	Platform as a Service (PaaS)
critical path	schedule performance index (SPI)
earned value (EV)	schedule variance (SV)
earned value management (EVM)	socioemotional leader
ecommerce project management	Software as a Service (SaaS)
economic feasibility	systems proposal
estimate at completion (EAC)	tangible benefits
estimate to complete (ETC)	tangible costs
expediting	task leader
forecasting	team motivation
function point analysis	team norms
Gantt chart	team process
Infrastructure as a Service (IaaS)	technical feasibility
Hardware as a Service (HaaS)	vendor support
intangible benefits	work breakdown structure (WBS)

Review Questions

1. What are the five major project fundamentals?
2. List three ways to find out about problems or opportunities that might call for a systems solution.
3. List the five criteria for systems project selection.
4. Define *technical feasibility*.
5. Define *economic feasibility*.
6. Define *operational feasibility*.
7. List four criteria for evaluating system hardware.
8. What two main options do organizations have for acquiring or using computer hardware?
9. What does COTS stand for?
10. Compare the advantages and disadvantages of purchasing computer resources versus using cloud services.
11. List the criteria used by organizations as guidelines for vendor selection.
12. What does bring your own device (BYOD) mean for systems analysts?
13. What are the benefits of BYOD to an organization?
14. What are the benefits of BYOD to an employee?
15. What is the biggest drawback of BYOD for an organization?
16. How can cloud computing decisions be addressed at a strategic level?
17. Define *tangible costs* and *tangible benefits*. Give an example of each one.
18. Define *intangible costs* and *intangible benefits*. Give an example of each one.
19. List the different forecasting models available to a systems analyst and the conditions for choosing them.
20. When is calculating the payback period useful?
21. What are the three drawbacks of using the payback method?
22. What is work breakdown structure (WBS), and when should it be used?
23. What is a Gantt chart?
24. When is a PERT diagram useful for systems projects?
25. List three advantages of using a PERT diagram over a Gantt chart for scheduling systems projects.
26. Define the term *critical path*.
27. Which form of analysis would you use to “sell” a systems project to a manager who is only interested in how many widgets need to be sold before the project becomes profitable? Why?
28. What costs need to be estimated to prepare a budget?
29. Why is preparing a budget so important for a systems analyst who is managing a project?
30. List the approaches to estimating the time required for the completion of a task.

31. What does crash time mean when expediting a project?
32. What is earned value management (EVM)?
33. What are the four key measures in earned value management?
34. How is earned value management used to compare actual and planned project development?
35. List the two types of team leaders.
36. What is meant by a dysfunctional team norm?
37. What is meant by team process?
38. What are three reasons that goal setting seems to motivate systems analysis team members?
39. What are four ways in which ecommerce project management differs from traditional project management?
40. What elements are contained in a project charter?
41. For what purpose is a fishbone diagram used?
42. What are the three steps a systems analyst must follow to put together an effective systems proposal?
43. List the 10 main sections of a systems proposal.

Problems

1. Williwonk's Chocolates of St. Louis makes an assortment of chocolate candy and candy novelties. The company has six in-city stores, five stores in major metropolitan airports, and a small mail-order branch. Williwonk's has a small, computerized information system that tracks inventory in its plant, helps schedule production, and so on, but this system is not tied directly into any of its retail outlets. The mail-order system is handled manually, and any emailed order forms received through the website are answered via email by clerical personnel, with orders being forwarded to the appropriate department.

Recently, several Williwonk's stores experienced a rash of complaints from mail-order customers that the candy was spoiled upon arrival, that it did not arrive when promised, or that it never arrived; the company also received several letters complaining that candy in various airports tasted stale. Williwonk's has been selling a new, low-carb, dietetic form of chocolate made with sugar-free artificial sweetener. Sales have been brisk, but there have been problems shipping the wrong type of chocolate to an address for a diabetic person. There were a number of complaints, and Williwonk's sent a number of free boxes of chocolate to ease the situation.

Management would like to sell products online but only has a basic website with information about the company and an order form that can be downloaded, printed, filled out by the customer, and returned via email. Web ordering does not exist. One of the senior executives would like to sell customized chocolates with the name of a person on each piece. Although the production area has assured management that this could be easily done, there is no method to order customized chocolates.

Another senior executive has mentioned that Williwonk's has partnered with several European chocolate manufacturers and will be importing chocolate from a variety of countries. At present, this must be done over the phone, by email, or by mail. The executive wants an internal, Internet-based network that will enable employees to order directly from the partner companies. All this has led a number of managers to request trend analysis. Too much inventory results in stale chocolate, and at other times there is a shortage of a certain kind of chocolate.

Seasonal and holiday variation trends would help Williwonk's maintain an adequate inventory. The inventory control manager has insisted that all changes must be implemented before the next holiday season. "The time for this to be complete is an absolute due date," remarked Candy, a senior manager. "Make sure that everything works perfectly before the site goes public," she continues. "I don't want any customers receiving the wrong chocolates!" In addition, the order processing manager has mentioned that the system must be secure.

You have been working for two weeks with Williwonk's on some minor modifications for its inventory information system when you overhear two managers discussing these occurrences. List the possible opportunities or problems among them that might lend themselves to systems projects.

2. Where is most of the feedback on problems with Williwonk's products coming from in Problem 1? How reliable are the sources? Explain in a paragraph.
3. After getting to know Williwonk's management people better, you have approached them with your ideas on possible systems improvements that could address some of the problems or opportunities given in Problem 1.
 - a. In two paragraphs, provide your suggestions for systems projects. Make any realistic assumptions necessary.
 - b. Are there any problems or opportunities discussed in Problem 1 that are not suitable? Explain your response.

4. Create a problem definition for Williwonk's, as described in Problem 1. Estimate the weights of importance. Include at least one requirement and one constraint.
5. Create a list of user requirements for the problem definition created in Problem 4.
6. Delicato, Inc., a manufacturer of precise measuring instruments for scientific purposes, has presented you with a list of attributes that its managers think are probably important in selecting a vendor for computer hardware and software, or a cloud service provider. The criteria are listed here, but not in order of importance:
 1. Low price
 2. Precisely written software for engineering applications
 3. Vendor performs routine maintenance on hardware
 4. Training for Delicato employees
 - a. Critique the list of attributes in a paragraph.
 - b. Using its initial input, help Delicato, Inc. draw up a more suitable list of criteria for selecting computer hardware and software vendors to purchase from.
 - c. Using its initial input, help Delicato, Inc. draw up a more suitable list of criteria for selecting cloud vendors who could provide HaaS and SaaS.
 - d. In a paragraph, state how the list of attributes for selecting a computer hardware vendor to purchase from should differ from the list of attributes for selecting a cloud provider for HaaS.
 - e. In a paragraph, state how the list of attributes for selecting a computer software vendor to purchase applications from should differ from the list of attributes for selecting a cloud provider for SaaS.
7. Pinacle, a mobile application development firm, is planning to develop a mobile application to search for cricket coaches. The application will help parents and students identify coaches available in the vicinity, check their profiles, coaching experience, fees, and timings. They can then book an appointment with the shortlisted coaches and choose the one who suits them best. The application is being designed for a single city but will be expanded to state-wide coverage.
 - a. Identify the company's likely hardware and software requirements for developing the mobile application.
 - b. Which hardware and software approach would you recommend from the alternatives listed in the text and why?
8. The following is 12 years' data for the annual debt of a firm.

Year	Debt
2001	846,000
2002	1,720,000
2003	2,285,000
2004	96,000
2005	93,000
2006	2,247,203
2007	1,268,386
2008	1,631,809
2009	995,204
2010	646,000
2011	1,873,678
2012	1,470,000

- a. Graph the debt data for the firm.
 - b. Determine the linear trend for the firm using a three-year moving average.
9. Which trend is the best fit for the firm's debt data in Problem 8? Explain.
10. Interglobal Health Consultants has asked for your help in comparing its present computer system with a new one its board of directors would like to see implemented. The proposed system costs and present system costs are as follows:

Item	Proposed System Costs	Present System Costs
Year 1		
Equipment lease	\$20,000	\$11,500
Salaries	30,000	50,000
Overhead	4,000	3,000
Development	30,000	—
Year 2		
Equipment lease	\$20,000	\$10,500
Salaries	33,000	55,000
Overhead	4,400	3,300
Development	12,000	—
Year 3		
Equipment lease	\$20,000	\$10,500
Salaries	36,000	60,000
Overhead	4,900	3,600
Development	—	—
Year 4		
Equipment lease	\$20,000	\$10,500
Salaries	39,000	66,000
Overhead	5,500	4,000
Development	—	—

Graph the costs of the proposed and present systems and show the break-even point.

11. The following are system benefits for Interglobal Health Consultants (from Problem 10):

Year	Benefits
1	\$55,000
2	75,000
3	80,000
4	85,000

Use the costs of Interglobal Global Health Consultants proposed system from Problem 10 to determine the payback period. (Use the payback method.)

12. Brian F. O'Byrne ("F," he says, stands for "frozen") owns a frozen foods company and wants to develop an information system for tracking shipments to warehouses.
- Using the data from the table in Figure 3.EX1, draw a Gantt chart to help Brian organize his design project.
 - When is it appropriate to use a Gantt chart? What are the disadvantages of using a Gantt chart? Explain in a paragraph.

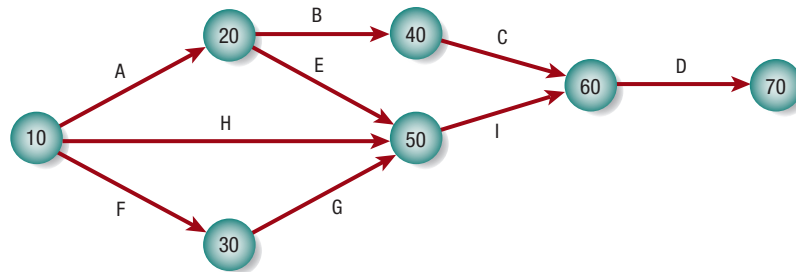
Description	Task	Must Follow	Time (Weeks)
Draw data flow	A	None	5
Draw decision tree	B	A	4
Revise tree	C	B	10
Write up project	D	C, I	4
Organize data dictionary	E	A	7
Do output prototype	F	None	2
Revise output design	G	F	9
Write use cases	H	None	10
Design database	I	H, E, and G	8

FIGURE 3.EX1

Data to help in the organization of a design project for creating an information system that tracks shipments of frozen foods to warehouses.

FIGURE 3.EX2

The PERT diagram for Brian's frozen foods company.



13. In addition to using a Gantt chart, you've drawn a PERT diagram for Brian so that you can communicate the necessity of keeping an eye on the critical path. Consult Figure 3.EX2, which was derived from the data from Figure 3.EX1. List all paths and calculate and identify the critical path.
14. Cherry Jones owns a homeopathic medicine company called Faithhealers. She sells vitamins and other relatively nonperishable products for those who want choices regarding alternative medicine. Cherry is developing a new system that would require her staff to be retrained. Given the information in Figure 3.EX3, make a PERT diagram for her and identify the critical path. If Cherry could find a way to save time on the "write use cases" phase, would it help? Why or why not?
15. Using the PERT diagram in Figure 3.EX2 to determine answers:
 - a. What activities can you expedite to complete the project a week ahead of schedule?
 - b. Suppose activity E is the least costly activity to expedite. What happens if you try to expedite the project by more than one week? Explain.
16. You can help Cherry finish her project faster by expediting. Here are the costs.

Activity	Estimated Duration	Crash Time	Cost per Week
A	6	3	600
B	3	2	100
C	2	2	400
D	3	2	400
E	4	2	200
F	2	2	200
G	5	3	300
H	3	2	400
I	2	2	300

If Cherry can keep her expediting expenses to \$325 per week, she would benefit from expediting because she can start a new project earlier than planned. If she goes over the \$325 per week limit, it would not be worthwhile.

- a. Which three activities cannot be expedited at all because they are at their crash times already?
- b. Which two activities cannot be expedited because their expediting cost exceeds the daily limit?
- c. Which two activities are unlikely candidates for expediting because they are not on the critical path?
- d. Set up a table and show step by step which activities should be expedited to shorten the project time. What is the minimum time the project will take if Cherry expedites the project as much as she can and stays within her limits?
- e. Why is Cherry's limit exceeded if you try to shorten the project by one week more?

FIGURE 3.EX3

Tasks to be performed during systems development of an order fulfillment system.

Description	Task	Must Follow	Time (Weeks)
Interview executives	A	None	6
Interview staff in order fulfillment	B	None	3
Design input prototype	C	B	2
Design output prototype	D	A, C	3
Write use cases	E	A, C	4
Record staff reactions to prototypes	F	D	2
Develop system	G	E, F	5
Write up training manual	H	B, G	3
Train staff working in order fulfillment	I	H	2

17. Robyn Cucurullo is developing a social networking app for tablet computers. She has:

Week or Stage	Estimated Cost of Stage	Stage Completed	Actual Cost of Stage to Date
Week 1	\$500	100%	500
Week 2	400	100%	500
Week 3	600	100%	700
Week 4	500	50%	50
Week 5	400	0	Not yet begun

- Create a table similar to the one in Figure 3.24.
 - Calculate the budget at completion (BAC), planned value (PV), actual cost (AC), and earned value (EV) at Week 3.
 - For Week 3, calculate the performance measures of cost variance (CV), schedule variance (SV), the cost performance index (CPI), and the schedule performance index (SPI).
 - What can you tell about the *budget* in Week 3 by using these performance measures?
 - What can you tell about the *schedule* in Week 3 by using these performance measures?
 - For Week 3, calculate the estimated time to complete (ETC) and estimate at completion (EAC).
18. Using the data in the table from the previous problem:
- Create a table similar to the one in Figure 3.24 if you haven't done so already.
 - Calculate the budget at completion (BAC), planned value (PV), actual cost (AC), and earned value (EV) at Week 4.
 - For Week 4, calculate the performance measures of cost variance (CV), schedule variance (SV), the cost performance index (CPI), and the schedule performance index (SPI).
 - What can you say about the *budget* in Week 4 by using these performance measures?
 - What can you say about the *schedule* in Week 4 by using these performance measures?
 - For Week 4, calculate the estimated time to complete (ETC) and estimate at completion (EAC).
19. Angus McIndoe wants to modernize his popular restaurant by adapting it more closely to the preferences of his repeat customers. This would involve keeping track of his customers' likes and dislikes. Information such as where they like to sit, what they like to eat, and when they normally arrive at the restaurant are all items of interest to Angus because he believes this information will help him better serve his customers. Angus has asked you to develop a system for him that will help make his customers happy while increasing his business.
- You have heard what Angus has to say about his customers. He can certainly keep track of additional preferences. Develop a problem definition for Angus similar to the one developed for Catherine's Catering in this chapter.
20. Recently, two analysts just out of college have joined your systems analysis group at the newly formed company Mega Phone. When talking to you about the group, they mention that some things strike them as odd. One is that group members seem to look up to two group leaders, Bill and Penny, not just one.
- Their observation is that Bill seems pretty relaxed, whereas Penny is always planning and scheduling activities. They have also observed that everyone "just seems to know what to do" when they get into a meeting, even though no instructions are given. Finally, they have remarked on the openness of the group in addressing problems as they arise, instead of letting things get out of hand.
- By way of explanation to the new team members, label the types of leaders Bill and Penny appear to be, respectively.
 - Explain the statement that "everyone just seems to know what to do." What is guiding their behavior?
 - What concept best describes the openness of the group that the new team members commented upon?
21. "I think it's only fair to write up *all* the alternatives you've considered," says Lou Cite, a personnel supervisor for DayGlow Paints. "After all, you've been working on this systems thing for a while now, and I think my boss and everyone else would be interested to see what you've found out." You are talking with Lou as you prepare to put together the final systems proposal that your team will be presenting to upper management.
- In a paragraph, explain to Lou why your proposal will not (and should not) contain all the alternatives that your team has considered.
 - In a paragraph, discuss the sorts of alternatives that should appear in the final systems proposal.

Group Projects

1. The Weil Smile Clinic is a dental practice run by Drs. Bonnie and Jeff, and they need to keep the necessary patient and insurance data safe and secure. They looked into online, cloud-based backups: SOS Online Backup, Spare Backup, MozyPro, and KineticD. With your group, research the cost of these or other services, and help Drs. Bonnie and Jeff make a decision. What are the intangible costs and benefits of backing up this way? Should they use a backup service or find some other way? Write two paragraphs defending your group's analysis and recommendations.
2. Explore four or five Voice over IP (VoIP) providers with your group.
 - a. Make a list of costs, including the setup fee, monthly cost of the basic plan, monthly cost of the unlimited plan, and cost of an adapter or other fees, if required.
 - b. Make a list of attributes, such as free in-network calls, international calling, virtual telephone numbers, teleconferencing, support for caller ID, and so on.
 - c. Write a paragraph with your group to explain how a person would use all the quantitative and qualitative information to make an informed decision about which VoIP provider to select. Are any other variables important? What type of software could be used to help compare these services?
3. With your group, choose one VoIP provider you would recommend to a client based on the analysis in Group Project 2, Parts a and b.
4. With your group members, explore project management software such as Microsoft Project.
 - a. What features are available? Work with your group to list them.
 - b. Have your group evaluate the usefulness of the software for managing a systems analysis and design team project. In a paragraph, state whether the software you are evaluating facilitates team member communication and management of the team, specifically: evaluate its capacity to coordinate activities, time, and resources. Also state which particular features support these aspects of any systems project. Also note whether the software falls short of these criteria in any regard.
5. A hospital is facing a number of issues in supply chain management (SCM), including overstocking, demand–supply gap, and transparency problems. In a group, brainstorm other possible issues that the hospital might be facing in SCM.
 - a. Prepare a list of the problems identified and categorize them appropriately.
 - b. Using your assumptions, draw a fishbone diagram to depict the identified problems.

Selected Bibliography

- Alter, S. *Information Systems: The Foundation of E-Business*, 4th ed. Upper Saddle River, NJ: Prentice Hall, 2002.
- Bales, R. F. *Personality and Interpersonal Behavior*. New York, NY: Holt, Rinehart and Winston, 1970.
- Bergstein, B. "IBM Faces the Perils of 'Bring Your Own Device.'" <https://www.technologyreview.com/s/427790/ibm-faces-the-perils-of-bring-your-own-device>. Last accessed August 23, 2017.
- Brownsword, L., C. Sledge, and P. Oberndorf. "An Activity Framework for COTS-Based Systems," (CMU/SEI-2000-TR-010), Software Engineering Institute, Carnegie Mellon University. <http://resources.sei.cmu.edu/library/asset-view.cfm?assetID=5225>. Last accessed August 23, 2017.
- Cearley, D. W. (2010). "Cloud Computing: Key Initiative Overview." https://www.gartner.com/it/initiatives/pdf/KeyInitiativeOverview_CloudComputing.pdf. Last accessed August 23, 2017.
- Construx Software Builders website. www.construx.com. Last accessed August 23, 2017.
- Costar website. www.softstarsystems.com. Last accessed July 28, 2012.
- Fisher, M. "What Happens When Work Becomes a Nonstop Chat Room." *New York Magazine*, May 1, 2017. <http://nymag.com/selectall/2017/05/what-has-slack-done-to-the-office.html>. Last accessed August 23, 2017.
- Fry, J. "Survey: Cloud Really Is Shaking Up the IT Role—With Some New Job Titles to Prove It." <http://datacenterdialog.blogspot.com/2011/03/survey-cloud-really-is-shaking-up-it.html>. Last accessed July 28, 2012. <http://www.get.slack.help>. Last accessed August 23, 2017.
- Germonprez, M., J. Kendall, K. Kendall, B. Warner, and L. Mathiassen. "Organizational Participation in Open Communities: Conceptual Framing and Early Findings." http://aisel.aisnet.org/amcis2011_submissions/242. Last accessed August 23, 2017.
- Glass, R. "Evolving a New Theory of Project Success." *Communications of the ACM* 42, 11 (1999): 17–19.
- Kaneshige, T. "The BYOD Sea Change Has Already Started." www.cio.com/article/701567/The_BYOD_Sea_Change_Has_Already_Started. Last accessed August 23, 2017.
- Levine, D. M., P. R. Ramsey, and M. L. Berenson. *Business Statistics for Quality and Productivity*. Upper Saddle River, NJ: Prentice Hall, 1995.

- Linberg, K. R. "Software Perceptions About Software Project Failure: A Case Study." *Journal of Systems and Software* 49, 2/3 (1999): 177–192.
- Linthicum, D. "3 Easy Steps to Creating Your Cloud Strategy." www.infoworld.com/d/cloud-computing/3-easy-steps-creating-your-cloud-strategy-193593. Last accessed August 23, 2017.
- Longstreet Consulting website. www.ifpug.org. Last accessed August 23, 2017.
- McBreen, P. *Questioning Extreme Programming*. Boston, MA: Addison-Wesley, 2003.
- Moyse, I. "Why More Education Is Needed on Cloud." www.cloudcomputing-news.net/news/2012/mar/16/employing-cloud/. Last accessed August 23, 2017.
- Moyse, I. "How Consumerisation Is Driving Cloud Acceptance." www.cloudcomputing-news.net/news/2012/mar/13/how-consumerisation-driving-cloud-acceptance. Last accessed August 23, 2017.
- Nash, K. S. "Cloud Computing: What CIOs Need to Know About Integration." <http://www.cio.com/article/2418286/software-as-a-service/cloud-computing—what-cios-need-to-know-about-integration.html>. Last accessed August 23, 2017.
- NASA. "NASA Open Source Software." <http://ti.arc.nasa.gov/opensource/>. Last accessed August 23, 2017.
- Peters, M. "How Cloud Computing and Web Services Are Changing the IT Job Market." <http://mashable.com/2011/02/26/it-job-market/>. Last accessed August 23, 2017.
- Schein, E. H. *Process Consultation: Its Role in Organization Development*. Reading, MA: Addison-Wesley, 1969.
- Shtub, A., J. F. Bard, and S. Globerson. *Project Management: Processes, Methodologies, and Economics*, 3d ed. Upper Saddle River, NJ: Pearson, 2005.
- Software Advice website. <http://www.softwareadvice.com/ca/erp/>. Last accessed August 23, 2017.
- Stefik, M., G. Foster, D. G. Bobrow, K. Kahn, S. Lanning, and L. Suchman. "Beyond the Chalkboard: Computer Support for Collaboration and Problem Solving in Meetings." *Communications of the ACM* 30, 1 (January 1987): 32–47.
- The Linux Foundation website. www.linuxfoundation.org. Last accessed August 23, 2017.
- Vigder, M. R., W. M. Gentleman, and J. C. Dean. "COTS Software Integration: State of the Art." NRC-CNRC Report, National Research Council, Canada, January 1996.
- Walsh, B. "Your Network's Not Ready for E-Commerce." *Network Computing*, 9, 22 (December 1998): 25–26.
- Yang, H., and M. Tate. "A Descriptive Literature Review and Classification of Cloud Computing Research." *Communications of the Association for Information Systems* 31, 2 (May 2012). <http://aisel.aisnet.org/cais/vol31/iss1/2>. Last accessed August 23, 2017.